EEBus UC Technical Specification

Monitoring of Battery

Version 1.0.0

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1 Scope of the document

This document describes the Use Case "Monitoring of Battery" (short-name: MOB). Chapter 2 specifies the High-Level Use Case. Chapter 3 details the technical solution for SPINE for this Use Case. Within this document, a top-down approach is used to derive the requirements for the technical solution from the High-Level description.

1.1 References

1.1.1 EEBUS documents

[ProtocolSpecification]	EEBus_SPINE_TS_ProtocolSpecification.pdf
[ResourceSpecification]	EEBus_SPINE_TS_ResourceSpecification.pdf
[SHIP]	EEBus_SHIP_TS_Specification_v1.0.1.pdf

1.1.2 Normative references

[RFC2119]	IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels Please see section 1.3.1 for details.

1.2 Terms and definitions

Actor

An Actor models a role within a Use Case definition (e.g. an energy manager or an electric vehicle).

AC

Abbreviation for alternating current.

Active sign convention

An electrical current is positive if the current is flowing out of device or component. In this case, the device or component produces electrical power and the active power is greater than zero. An electrical current is negative if the current is flowing into a device or component. In this case, the device or component consumes electrical power and the active power is smaller than zero.

CEM

Abbreviation for Customer Energy Manager. The CEM is an energy manager located at the user's home or premises or in a cloud application. The energy manager enables energy-optimized operation of the connected devices by harmonizing energy demand and availability.

DC

Abbreviation for direct current.

EV

Abbreviation for electric vehicle.

HVAC

Abbreviation for heating, ventilation and air conditioning.

MOB

Monitoring of Battery (short-name of this Use Case)

MOI

Short name of the Use Case "Monitoring of Inverter"

Monitoring Appliance

The Actor Monitoring Appliance evaluates particular data of another Actor.

Passive sign convention

An electrical current is positive if the current is flowing into a device or component. In this case, the device or component consumes electrical power and the active power is greater than zero. An electrical current is negative if the current is flowing out of a device or component. In this case, the device or component produces electrical power and the active power is smaller than zero.

PV

Abbreviation for photovoltaic.

PV String

Several PV modules are connected in groups parallel to the inverter. A group of modules connected in series is called a PV string.

Scenario

Part of a Use Case. Splitting a Use Case into Scenarios helps to understand the Use Case more quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or optional.

Specialization

Reusable data collection for a specific functionality.

SPINE

Smart Premises Interoperable Neutral-message Exchange: Technical Specification of EEBus Initiative e.V.

UI

Abbreviation for user interface.

UTC

Abbreviation for Coordinated Universal Time.

1.3 Requirements

1.3.1 Requirements wording

The following keywords are used:

- SHALL
- SHALL NOT
- SHOULD
- SHOULD NOT

- MAY

Note: They apply only if written in capital letters.

For the meaning of the keywords, please refer to [RFC2119].

1.3.2 Mapping of High-Level requirements

Within the High-Level Use Case description, the following abbreviation is used:

[MOB-xyz]

e.g.: [MOB-007]

The abbreviation is used to mark High-Level requirements or rules of this Use Case with a unique number xyz. These requirements are referenced throughout the technical solution to show how each High-Level requirement is realized in the technical part.

2 High-Level description

2.1 Introduction

Electrical storage systems are essential to realize the energy transition, as they enable power generation to be independent of consumption. On the one hand the mobility transition and CO2 saving goal bring new large loads like electric vehicles (EVs) and heating, ventilation and air conditioning (HVAC) systems to buildings, on the other hand the energy transition leads into much more use of fluctuating energy like wind or photovoltaic (PV).

The growing share of electric vehicles further increases the need for electrical storage systems to enable recharging independent of the availability of electrical grid energy. Faster recharging of an EV or even nearly free of charge recharging through PV energy that was not power curtailed during the day will be enabled at reduction of load on the public electricity grid at the same time.

An electrical storage system (in the following called battery) is charged and discharged with direct current (DC). To integrate the battery into the local or public electricity grid, an inverter is needed that converts the direct current into alternating current (AC).

A client, e.g. a customer energy manager (CEM) or a user interface (UI), may monitor data such as identification information, the state of the battery, power/current/voltage, or nominal values.

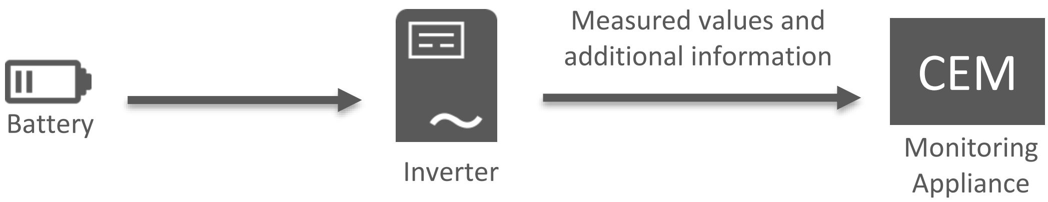


Figure 1: High-Level Use Case functionality overview

The retrieved data may be used for energy management purposes, debugging, or to just visualize the data to the customer. Some examples are given in the following list:

- Optimization of self-consumption
- Increase of autarky rate
- Appliance identification
- Efficiency calculation
- Error detection

2.2 Detailed background information

This Use Case describes a battery's data in context of a hybrid or battery inverter.

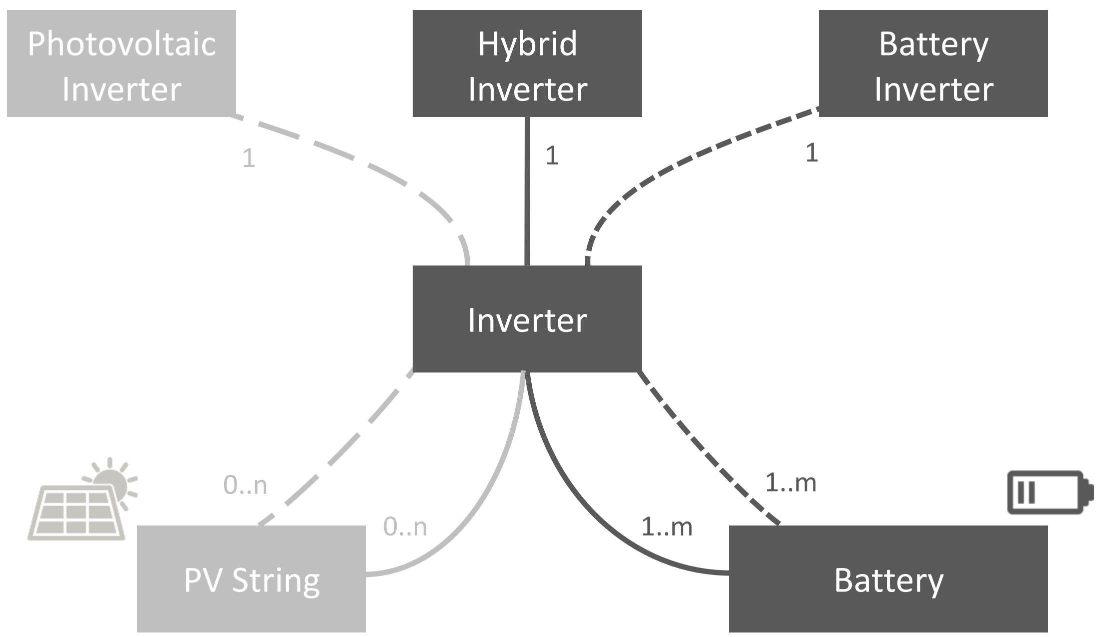


Figure 2: Example for Inverter device-hierarchy

The context of the Inverter is given by its so-called "parent" in the device-hierarchy ([MOB-004]). This can be:

- "Hybrid Inverter" ([MOB-004/1]), meaning that the Inverter is a hybrid Inverter that has both PV Strings (up to "n" in Figure 2) and Batteries (up to "m" in Figure 2) as "children" in its device hierarchy
- "Battery Inverter" ([MOB-004/2]), meaning that the Inverter has (at least) one Battery as "child" in its device hierarchy.
- Note: Other Use Cases may also specify a "Photovoltaic Inverter", where the Inverter has only PV Strings as so-called "children" in its device hierarchy. This type of Inverter is not considered in this Use Case.

Note: The "0..n" of Figure 2 just expresses that this Use Case does neither require nor describe data of any PV String itself, as such information is covered by PV String specific Use Cases. Instead, this Use Case covers information of an attached Battery.

2.3 Actors

2.3.1 Monitoring Appliance

The Actor Monitoring Appliance (e.g., a CEM) can manage the home's appliances to ensure energy-optimized operation of the appliances or visualize the information. In this Use Case, the Monitoring Appliance retrieves data provided by a Battery via a Hybrid or Battery Inverter.

2.3.2 Battery

The Battery is a sub-device of the Inverter. It represents one Battery that can store or release energy (in form of DC power).

2.4 Scenarios

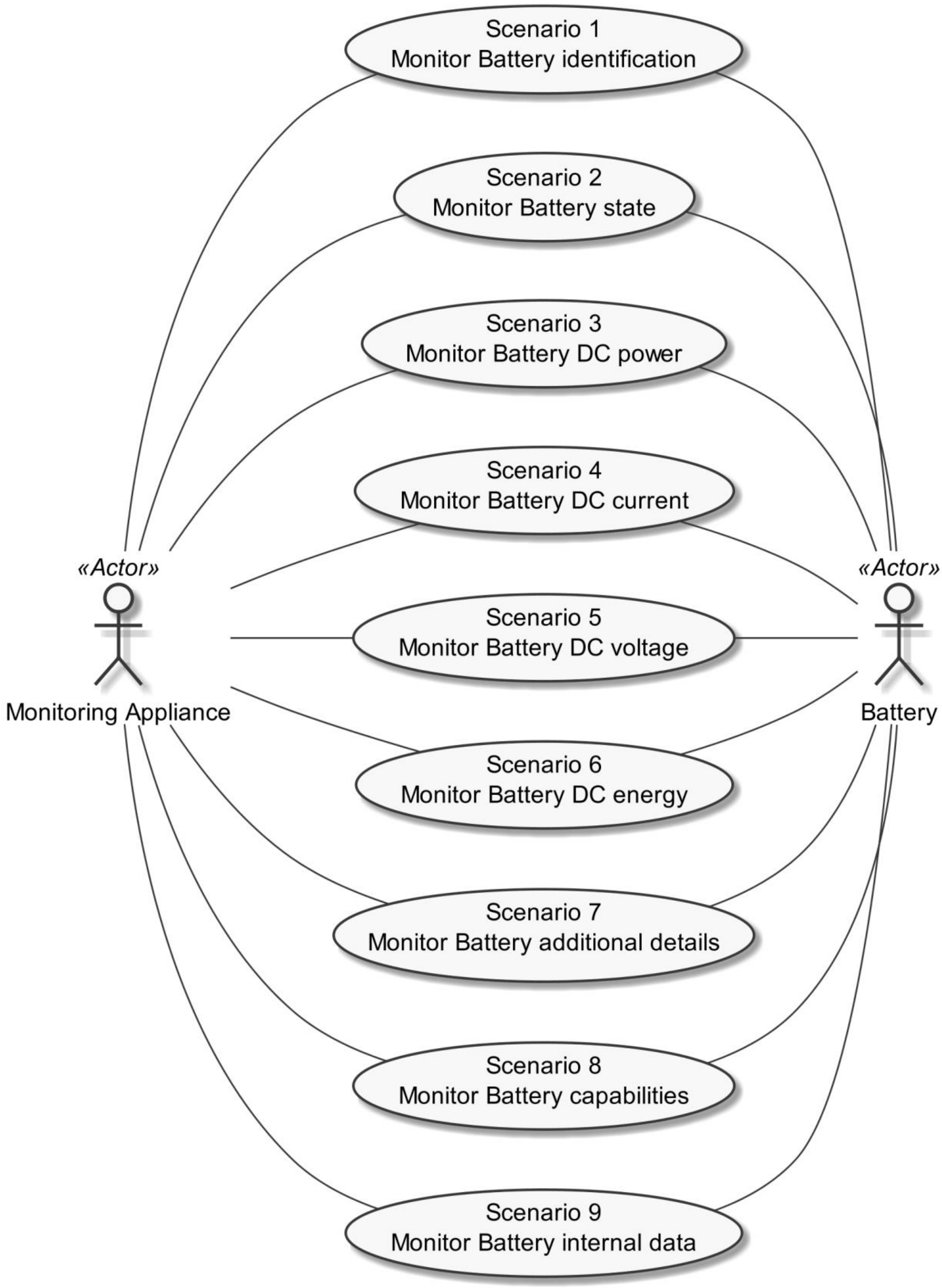


Figure 3: Scenario overview

Scenario number	Scenario name	Monitoring Appliance	Battery
1	Monitor Battery identification	M	R*[1]
2	Monitor Battery state	M	M
3	Monitor Battery DC power	R	M
4	Monitor Battery DC current	M	M
5	Monitor Battery DC voltage	M	M
6	Monitor Battery DC energy	M	R
7	Monitor Battery additional details	R	R
8	Monitor Battery capabilities	R	M
9	Monitor Battery internal data	O	O

Table 1: Scenario implementation requirements for Actors

*[1]: Scenario 1 ist not mandatory for the Actor Battery, as there might be installations where the Battery's parent device (see section 2.2) is not capable of providing identification information for the Battery.

2.4.1 Scenario 1 - Monitor Battery identification

2.4.1.1 Description

This Scenario is used to identify the Actor Battery. Depending on the information provided by the Inverter, several data points are available to be requested by the Actor Monitoring Appliance.

Data point name	Data point description	Support indication	High-Level requirement
Device Name	Name of the Battery part as indicated on the packaging.	R	[MOB-011]
Device Code	Code of the Battery part to uniquely identify the exact model.	R	[MOB-012]
Serial Number	Unique serial number of the Battery part.	R	[MOB-013]
Software Revision	Version of the software on the Battery part.	R	[MOB-014]
Hardware Revision	Hardware version of the Battery part.	R	[MOB-015]
Manufacturer Name	Name of the manufacturer of the Battery part.	M	[MOB-016]
Battery Label	Label for the Battery part (e.g. for display purposes).	O	[MOB-017]

Table 2: Scenario 1 - Monitor Battery identification - Data point list

2.4.1.2 Conditions

Triggering Event:

The Actor Monitoring Appliance connects to the Actor Battery.

Pre-condition:
The Actor Monitoring Appliance does not know the Battery's identification information.

Post-condition:
The Actor Monitoring Appliance knows the identification information provided by the Actor Battery.

2.4.2 Scenario 2 - Monitor Battery state

2.4.2.1 Description

The state values of the Actor Battery are provided by this Scenario. The following data points can be provided by the Battery:

Data point name	Data point description	Support indication	High-Level requirement
Battery State of Charge	Current state of charge in percentage of the "Battery Usable Capacity" ([MOB-024]).	M	[MOB-021]
Battery State of Health	Current state of health in percentage of "Battery Capacity Nominal Max" ([MOB-081]).	R	[MOB-022]
Battery State of Energy	Remaining energy of the Battery in Wh.	R	[MOB-023]
Battery Usable Capacity	Usable part of the capacity of the Battery in Wh.	R	[MOB-024]
Battery State	State of the Battery. Possible states: - Normal operation [MOB-025/1] - Standby [MOB-025/2] - Temporarily not ready [MOB-025/3] - Off [MOB-025/4] - Failure [MOB-025/5]	M	[MOB-025]

Table 3: Scenario 2 - Monitor Battery state - Data point list

This specification defines the following relations:

1. "Battery Usable Capacity" ([MOB-024]) = "Battery Capacity Nominal Max" ([MOB-081]) * "Battery State of Health" ([MOB-022])
2. "Battery State of Energy" ([MOB-023]) = "Battery Usable Capacity" ([MOB-024]) * "Battery State of Charge" ([MOB-021])

Note: The data points and their relations describe ideal conditions. In practice, external conditions (temperature or magnitude of a discharge power, e.g.) as well as internal conditions (intrinsic limitations on measurement accuracy, curtailment of discharge level, etc.) may lead to inaccuracies.

2.4.2.2 Conditions

Triggering Event:
The Actor Monitoring Appliance is interested in the state values of the Actor Battery.

Pre-condition:
The Actor Monitoring Appliance does not know the state values of the Actor Battery.

Post-condition:
The Actor Monitoring Appliance knows the state values of the Actor Battery.

2.4.3 Scenario 3 - Monitor Battery DC power

2.4.3.1 Description

The DC power values of the Actor Battery are provided by this Scenario. The following data points can be provided by the Battery:

Data point name	Data point description	Support indication	High-Level requirement
Battery DC Power	DC power of the Battery.	M	[MOB-031]

Table 4: Scenario 3 - Monitor Battery DC power - Data point list

2.4.3.2 Conditions

Triggering Event:
The Actor Monitoring Appliance is interested in the DC power values of the Actor Battery.

Pre-condition:
The Actor Monitoring Appliance does not know the DC power values of the Actor Battery.

Post-condition:
The Actor Monitoring Appliance knows the DC power values of the Actor Battery.

2.4.4 Scenario 4 - Monitor Battery DC current

2.4.4.1 Description

The DC current values of the Actor Battery are provided by this Scenario. The following data points can be provided by the Battery:

Data point name	Data point description	Support indication	High-Level requirement
Battery DC Current	DC current of the Battery.	M	[MOB-041]

Table 5: Scenario 4 - Monitor Battery DC current - Data point list

2.4.4.2 Conditions

Triggering Event:
The Actor Monitoring Appliance is interested in the DC current values of the Actor Battery.

Pre-condition:
The Actor Monitoring Appliance does not know the DC current values of the Actor Battery.

Post-condition:
The Actor Monitoring Appliance knows the DC current values of the Actor Battery.

2.4.5 Scenario 5 - Monitor Battery DC voltage

2.4.5.1 Description

The DC voltage values of the Actor Battery are provided by this Scenario. The following data points can be provided by the Battery:

Data point name	Data point description	Support indication	High-Level requirement
Battery DC Voltage	DC voltage of the Battery.	M	[MOB-051]

Table 6: Scenario 5 - Monitor Battery DC voltage - Data point list

2.4.5.2 Conditions

Triggering Event:
The Actor Monitoring Appliance is interested in the DC voltage values of the Actor Battery.

Pre-condition:
The Actor Monitoring Appliance does not know the DC voltage values of the Actor Battery.

Post-condition:
The Actor Monitoring Appliance knows the DC voltage values of the Actor Battery.

2.4.6 Scenario 6 - Monitor Battery DC energy

2.4.6.1 Description

The DC energy values of the Actor Battery are provided by this Scenario. The following data points can be provided by the Battery:

Data point name	Data point description	Support indication	High-Level requirement
Battery Total DC Charge Energy	Total amount of energy that was charged into the Battery. Accumulated over lifetime or since last reset.	M	[MOB-061]
Battery Total DC Discharge Energy	Total amount of energy that was discharged from the Battery. Accumulated over lifetime or since last reset.	M	[MOB-062]

Table 7: Scenario 6 - Monitor Battery DC energy - Data point list

2.4.6.2 Conditions

Triggering Event:
The Actor Monitoring Appliance is interested in the DC energy values of the Actor Battery.

Pre-condition:
The Actor Monitoring Appliance does not know the DC energy values of the Actor Battery.

Post-condition:
The Actor Monitoring Appliance knows the DC energy values of the Actor Battery.

2.4.7 Scenario 7 - Monitor Battery additional details

2.4.7.1 Description

Some additional values of the Actor Battery are provided by this Scenario. The following data points are specified:

Data point name	Data point description	Support indication	High-Level requirement
Cumulated Load Cycle Count	Number of full charge/discharge cycles*[1] of the Battery since installation (or last reset).	O	[MOB-071]
Battery Type	Type of the Battery. Possible types are: - Lithium-ion [MOB-072/1] - Lead-acid [MOB-072/2] - Redox flow [MOB-072/3] - Sodium-ion [MOB-072/4]	O	[MOB-072]

Table 8: Scenario 7 - Battery additional details - Data point list

*[1]: Even though the Cumulated Load Cycle Count is an integer value, the Inverter should internally count partial charge/discharge cycles with an according fraction. This is a vendor specific implementation.

Note: If this Scenario is supported, at least one of the values stated above SHALL be supported.

2.4.7.2 Conditions

Triggering Event:
The Actor Monitoring Appliance is interested in the additional detail values of the Actor Battery.

Pre-condition:
The Actor Monitoring Appliance does not know about the additional detail values of the Actor Battery.

Post-condition:
The Actor Monitoring Appliance knows the additional detail values of the Actor Battery.

2.4.8 Scenario 8 - Monitor Battery capabilities

2.4.8.1 Description

The capability information of the Actor Battery is provided by this Scenario. The following data points can be provided by the Battery:

Data point name	Data point description	Support indication	High-Level requirement
Battery Capacity Nominal Max	The nominal capacity of the Battery in Wh as specified in the manufacturer's data sheet. For the usable part of the capacity of the Battery see [MOB-024].	M	[MOB-081]
Battery DC Charge Power Nominal Max	The nominal maximum DC power the Battery can be charged with, as specified in the manufacturer's data sheet. For the measured value see [MOB-031].	R	[MOB-082]
Battery DC Discharge Power Nominal Max	The nominal maximum DC power the Battery can be discharged with, as specified in the manufacturer's data sheet. For the measured value see [MOB-031].	R	[MOB-083]
Battery Maximum Charge/Discharge Cycle Count Per Day	Maximum number of cycles the battery permits per day.	O	[MOB-084]

Table 9: Scenario 8 - Monitor Battery capabilities - Data point list

2.4.8.2 Conditions

Triggering Event:
The Actor Monitoring Appliance is interested in the capability information of the Actor Battery.

Pre-condition:
The Actor Monitoring Appliance does not know the capability information of the Actor Battery.

Post-condition:
The Actor Monitoring Appliance knows the capability information of the Actor Battery.

2.4.9 Scenario 9 - Monitor Battery internal data

2.4.9.1 Description

Some internal data of the Actor Battery are provided by this Scenario. The following data points are specified:

Data point name	Data point description	Support indication	High-Level requirement
Internal Temperature	Internal component temperature. It is not specified further where the temperature sensor is located within the Actor.	O	[MOB-091]

Table 10: Scenario 9 - Monitor Battery internal data - Data point list

Note: If this Scenario is supported, at least one of the values stated above SHALL be supported.

2.4.9.2 Conditions

Triggering Event:
The Actor Monitoring Appliance is interested in the internal data of the Actor Battery.

Pre-condition:
The Actor Monitoring Appliance does not know the internal data of the Actor Battery.

Post-condition:
The Actor Monitoring Appliance knows the internal data of the Actor Battery.

2.5 Dependencies to other Use Cases

2.5.1 "Monitoring of Inverter"

The Use Case "Monitoring of Inverter" (MOI) describes the information provided by the Actor Inverter.

The Actor Inverter as parent Actor of the Actor Battery of this Use Case (see section 2.2) SHALL support the Use Case "Monitoring of Inverter".

The Actor Monitoring Appliance SHALL support the Use Case "Monitoring of Inverter".

2.6 Further information and rules

2.6.1 Sign conventions

Throughout the whole Use Case the "load convention" (i.e. "passive sign convention") is applied for measurement values ([MOB-001]). This means measured electrical current, active power and exchanged energy are expressed with positive values in case of energy consumption whereas negative values are used in case of energy production. This holds valid for the following values of this Use Case:

- Battery DC Power ([MOB-031])
- Battery DC Current ([MOB-041])
- Battery Total DC Charge Energy ([MOB-061])
- Battery Total DC Discharge Energy ([MOB-062])

The Battery's voltage ([MOB-051]) is measured independent of the energy direction ([MOB-002/1]).

Percentage values are always expressed as non-negative value between 0 and 100 ([MOB-002/2]). This holds valid for the following values of this Use Case:

- Battery State of Charge ([MOB-021])
- Battery State of Health ([MOB-022])

The data points that indicate energy capacities of the Battery are always expressed as non-negative value ([MOB-002/3]). This holds valid for the following values of this Use Case:

- Battery State of Energy ([MOB-023])
- Battery Usable Capacity ([MOB-024])
- Battery Capacity Nominal Max ([MOB-081])

The sign convention described above does NOT apply to nominal values. For nominal values that are dedicated to a particular energy direction it is common practice to express them as positive value. Their name indicates whether they are used for consumption or production direction ([MOB-003]):

- Battery DC Charge Power Nominal Max ([MOB-082]): consumption direction
- Battery DC Discharge Power Nominal Max ([MOB-083]): production direction

2.6.2 Value state rules

Each Battery's measurement value has a state that defines whether the value is correct or is to be ignored by the Monitoring Appliance ([MOB-005]):

- normal: value correct
- out of range: value is out of range and SHALL be ignored by the Monitoring Appliance
- error: value is erroneous and SHALL be ignored by the Monitoring Appliance.

2.7 Assumptions and Prerequisites

None.

3 Technical SPINE solution

3.1 General rules and information

3.1.1 Underlying technology documents

This technical solution relies on the SPINE Resources Specification version 1.2.0 [ResourceSpecification].

For interoperable connectivity this technical solution relies on:

- SPINE Protocol Specification version 1.2.0 [ProtocolSpecification] as application protocol.
- SHIP Specification version 1.0.1 [SHIP] as transport protocol.

3.1.2 Use Case discovery rules

Use Case discovery SHALL be supported by each Actor and the following rules SHALL apply:

- The string content for the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseName" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be "monitoringOfBattery". The string content SHALL only be defined by this Use Case (regardless of the Use Case version).
- The string content of the Element "nodeManagementUseCaseData. useCaseInformation. actor" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be set to the according value stated within the corresponding Actor's section.
- An Actor A that is implemented to support this Use Case specification SHALL set the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseVersion" within the Use Case discovery (please refer to [ProtocolSpecification]) to "1.0.0".
- If an Actor A supports multiple versions of this Use Case with the same major version number, only the highest one SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports multiple versions of this Use Case with the same major version number as supported by Actor A, the Actor A SHOULD evaluate from these versions of Actor B only the highest version number.
- If an Actor A supports multiple versions of this Use Case with different major version numbers, for each major version number only the highest version number SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports only versions with a major version number not implemented by Actor A, it still might be possible to run the Use Case or parts of the Use Case. Therefore, the Actor A should try to evaluate the Actor B as a valid partner for this Use Case.

3.1.3 Rules for "Content of Specialization..." tables and "Content of Function..." tables

3.1.3.1 General presence indication definitions

Abbreviations for the presence indication of Elements listed in the tables are defined as follows:

Abbreviation	Meaning	Link to requirement keywords

M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 11: Presence indication description

An Actor MAY support Elements that are not listed in the tables. However, another Actor MAY ignore these Elements.

The presence indications "M", "R" and "O" are always meant relative to the respective parent Element. I.e. if a parent Element is optional ("O") and a child is mandatory ("M") the child Element can only be present if the parent Element is present as well.

Note: The indications and the aforementioned rules apply for "complete messages" (so-called "full function exchange", please refer to [ProtocolSpecification]). In contrast, the so-called "restricted function exchange" is designed to permit exchange of specific excerpts of data, i.e. fewer Elements than potentially available from the data owner (partially even not all "mandatory" Elements).

3.1.3.2 Presence indications for "Content of Specialization..." tables

This section only defines rules for the client side.

Elements that are marked with "M" SHALL be supported by the client in case of readable as well as writeable data. This Element may be optional on the server side.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be supported by the client. In other words: If the server responds with the according Element, the client SHOULD be able to interpret the according Elements.
- "O" means that the data MAY be supported by the client. In other words: If the server responds with the according Element, the client MAY be able to interpret the according Elements.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be written by the client.
- "O" means that the data MAY be written by the client.
- "F" means that the data SHALL NOT be written by the client.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "reply" message: The client MAY ignore the Element.
- In case of a "write" operation to be created: The client MAY set the Element but SHALL consider that the server may ignore the Element.
- In case of a received "notify" message: The client MAY ignore the Element.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- M, O
- M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.3 Presence indications for "Content of Function..." tables

This section only defines rules for the server side.

Elements that are marked with "M" SHALL be supported by the server in case of readable as well as writeable data. In case of writeable data (marked with "M \W") the server does not need to set the Element, because the Element is set only by the client.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be provided by the server.
- "O" means that the data MAY be provided by the server.
- "F" means that the data SHALL NOT be provided by the server.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be supported. In other words: If the client writes the Element, the server SHOULD accept those messages and the contained Elements.
- "O" means that the data MAY be supported. In other words: If the client writes the Element, the server MAY accept those messages and the contained Elements.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "read" request: The according Element MAY be set in the reply.
- In case of a received "write" operation: The server MAY ignore the Element.
- In case of a "notify" operation to be created: The server MAY set the Element.

Note: The server will only accept write operations if the result fulfils the server Function requirements (permitted values, e.g.). Write operations on Elements that are not writeable MAY result in an error message.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- M, O
- M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.4 Cardinality indications on Elements and list items

A cardinality indication on an Element or list item expresses constraints on the number of occurrences of a given Element or data set. In this section we use "X" as representation for such an Element or data set. Furthermore, "a" and "b" represent constraints. The following rules apply for the occurrence of "X" and its content related to a specific Scenario (see note underneath the list):

1. X
   No cardinality indication.
2. X (a..b)
   This means "X" SHALL occur at least "a" times and at maximum "b" times.
3. X (a..)
   This means "X" SHALL occur at least "a" times and MAY occur more than "a" times.
4. X (..b)
   This means "X" SHALL occur at maximum "b" times and MAY occur less than "b" times (even zero occurrences are permissive).

Note: These rules apply only under consideration of presence indications and with regards to the given Scenario or Function definition for this Use Case.

The following table is an example to explain this for two different placements.

Scenario [{…}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
...	...		...
2: M \W	xFeatureType. xListData. xData. (1..3)		
2: M \W	xId	<*#(1..)>	PRIMARY IDENTIFIER
2: M \W	timePeriod		...
2: M \W	timePeriod. startTime	<xs:duration>	
2: M \W	xSlot. (1..)		
2: M \W	xSlot. xSlotId		...
2: M \W	xSlot. duration	<xs:duration>	...
...	...	...	...

Table 12: Example table for cardinality indications on Elements and list items

The field

xFeatureType. xListData. xData. (1..3)

introduces a data pattern (required Elements and values) for "xData" instances used for Scenario 2. The field itself specifies that such an "xData" instance SHALL occur at least 1 time and at maximum 3 times within "xListData" of Feature Type "xFeatureType". However, this constraint holds only for Scenario 2 and only if such "xData" are required. In this case, they are required, as the left field

2: M \W

denotes that this data set is mandatory for Scenario 2.

The field

xSlot. (1..)

expresses that the Element "xSlot" SHALL occur at least one time within its "xData", but MAY occur more than one time.

For the expression "<*#(1..)>" of Element "xId" please see section 3.1.3.6.

The remaining fields do not have an explicit cardinality indication.

Note: Cardinality expressions are also used within placeholder expressions as defined in section 3.1.3.6. In many cases such placeholder expressions define the number of required or permitted Elements or list items as they explicitly define how many different values for a given Identifier are required or permitted for a given Scenario.

3.1.3.5 Writability and changeability indication

In the same column where the presence indications are denoted, a mark is used to distinguish between writeable, changeable or readable Elements:

- Elements that are marked with "\W" are written by a client and SHALL be writeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\C" are changed by a client and SHALL be changeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\RW" are read and written by a client and SHALL be writeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are marked with "\RC" are read and changed by a client and SHALL be changeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are not marked are only read by a client and SHALL be provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.

"Writeable" means that the Element and its value may be written by a client. This includes the possibility to modify (if the Element is already present), create (if the Element is not present yet), and delete the Element. The server SHALL adjust its Function according to the received "write" operation (unless the server cannot accept the "write" operation according to section 3.1.3.3).

"Changeable" means that the Element's value may be changed by a client. If the Element is not present at the resource before, it probably **cannot** be created by the client via the "write" operation. In this case the server MAY decline such a message.

Note: "\W" includes "\C" already.

Note: Depending on the resource a client might need to request a proper binding before the server accepts a "write" operation.

3.1.3.6 "Value" placeholders

3.1.3.6.1 Introduction

Specializations may use placeholders to model relations between different Elements or even list items of different Functions. The main purpose is to declare which Identifier values relate to each other. As a Use Case does not prescribe specific values to be used for a given Identifier, a placeholder like "<x1>" can be used in "Value" columns to express the intended relations.

There are two styles to identify a placeholder that can be referenced:

1. <xM>
2. <xM#S>

where

1. "x" is any alphabetical prefix like "m", "t", "ec", "cc", etc.
2. "M" is a (major) number like "1", "2", "15", etc.
3. "S" is a sub-number like "1", "7", "10", etc.

Examples for the first style are "<ec1>", "<z12>". Examples for the second style are "<p4#2>", "<m22#3>". For a given placeholder, only one of the styles can be used.

In addition, there are also styles for placeholders that do not need to be referenced:

1. <*>
2. <*#S>

The second style is only used with so-called cardinality expressions.

3.1.3.6.2 Uniqueness of placeholders

A given placeholder <xM> or <xM#S> represents the same value throughout a given Use Case specification for a given set of its parent Identifier values. This shall be explained in a brief example:

We assume a list item with PRIMARY IDENTIFIER "pId". It also has a SUB IDENTIFIER "sId" with placeholder "<s1>". Then, each occurrence of "<s1>" represents the same value for a given value of pId. This means that "<s1>" of a list item with pId=1 can differ from "<s1>" of a list item with pId=2. But it also means that "<s1>" represents the same value in all list items with pId=1.

Note: Typically, parent Identifiers like "pId" will also be expressed with a placeholder like "<p5>", e.g. In this case, the uniqueness rule applies for "<p5>" likewise.

Note: The uniqueness also applies for placeholders used as FOREIGN IDENTIFIER.

3.1.3.6.3 Placeholder expressions with cardinalities

For some Identifiers, more than one placeholder is needed. Several notations are used for this purpose, which make use of cardinality expressions. The general notation is as follows:

1. <xM#(a..b)>

This is equivalent to this explicit definition:

At least a and at maximum b placeholders of this list: <xM#1> <xM#2> ... <xM#b>

This means that the implementation of a given Use Case (or Scenario) requires a minimum of "a" distinct values of the respective Identifier. In total, there can be up to "b" distinct values of the respective Identifier.

Additionally, the following notations may occur:

2. <xM#(a..)>
   This is equivalent to "<xM#(a..b)>" with "b" equal to infinity.
3. <xM#(..b)>
   This is equivalent to "<xM#(a..b)>" with "a" equal to zero.

"<xM#(a..)>" has only a lower bound of "a" distinct values, but no upper bound. "<xM#(..b)>", on the other hand, expresses that the Identifier may not be required at all, but it is permitted to have up to "b" distinct values.

Similarly, the cardinality can be used for placeholders that are not referenced, i.e. <*#(a..b)> etc.

Note: The cardinality does NOT express which of the sub-numbers have to be used! I.e., it does NOT mean that the Identifiers <xM#1> ... <xM#a> are always used and just those with larger sub-numbers (<xM#a+1> ... <xM#b>) are optional. If, for instance, a placeholder expression "<xM#(3..5)>" is given, it may well happen that an implementation makes use of <xM#2>, <xM#4>, and <xM#5> (i.e., it does NOT use <xM#1>, <xM#3>). Which sub-numbers are used usually depends on other parts of a Specialization and their references to required placeholders, which is explained in section 3.1.3.6.4.

3.1.3.6.4 References to placeholders and relations

According to the styles for placeholders that can be referenced, an enumeration value "e" can refer to a particular placeholder:

1. e(-><xM>)
2. e(-><xM#S>)

This denotes that "e" is to be used with "<xM>" or "<xM#S>", resp.

Example: A Specialization contains the Elements "mId" and "phase". "mId" is an Identifier with placeholder definition <m2#(1..3)>. "phase" is a string that permits the values "a", "b", and "c" using this expression:

"a"(-><m2#1>)
"b"(-><m2#2>)
"c"(-><m2#3>)

This expresses that the enumeration value "a" is to be used with the placeholder <m2#1>, "b" is to be used with <m2#2> and "c" with <m2#3>.

Similarly, a placeholder "yN" can refer to a particular placeholder:

3. <yN->xM>
4. <yN->xM#S>
5. <yN#T->xM>
6. <yN#T->xM#S>

where "T" is a sub-number of "yN".

It is also feasible to associate placeholders with cardinalities:

7. <yN#(a..b)->xM#(a..b)>

denotes that <yN#1> is to be used with <xM#1>, <yN#2> is to be used with <xM#2>, etc.

Note: In this case, the placeholder expressions of yN and xM must have the same cardinality.

In some cases, there is a need to express that multiple list items with similar values are feasible or required, but only particular combinations of these different data are permitted. The following example shows that several "fData" list items with different "phase" value are required, but that these list items may only express either the "phase" value combination { "a", "b", "c" } or the "phase" value combination { "a", "abc", "neutral" }. The permitted combinations are defined in a note below a table:

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	F. fListData. fData.		
2: M	zId	<z3#(3..5)>	
2: M	phase	"a"(-><z3#1>)	
		"b"(-><z3#2>)	
		"c"(-><z3#3>)	
		"abc"(-><z3#4>)	
		"neutral"(-><z3#5>)	

Table 13: Content of an example Specialization

Note: One of the following combinations SHALL be used at least: {<z3#1>, <z3#2>, <z3#3>} or {<z3#1>, <z3#4>, <z3#5>}.

3.1.3.7 Rules for content of "Value" column

For a given Scenario, the "Value" column may restrict the permitted content of a Function's Element to one or more particular values. This means that Elements with values deviating from the restriction (i.e. from the permitted values) do not belong to the respective Scenario and need to be considered as if the Element is not set. If more than one particular value is permitted for an Element, the values are in a single line each.

If a presence indication is set for the value (in an additional column before the value), the following rules SHALL be applied:

- "M" means that the value SHALL be supported. This means the value needs to be set at a certain point in time (depending on the value rules) or for a certain Element within a list entry.
- "R" means that the value SHOULD be supported.
- "O" means that the value MAY be supported.

If all possible values of a given mandatory Element are optional or recommended and this Element is used for the purpose of the respective Scenario, one of the values SHALL be set. If all possible values of a given optional or recommended Element are optional or recommended, this Element MAY contain also other values, but then this Element SHALL NOT be considered as part of the respective Scenario.

M, R or O may be combined with the suffix "(event)" to express that a supported value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

If no presence indication is set for the value, the following rules SHALL be applied:

- In case of Elements where the server may set or change an Element on its own (see section 3.1.3.5):
  - within the tables in the "Server data - Resources" sections:
    - the server SHALL support at least one of the listed values.
  - within the tables in the "Client data - Specializations" sections:
    - the client SHALL support all listed values.
- In case of Elements that are writeable or changeable (see section 3.1.3.5):
  - within the tables in the "Server data - Resources" sections:
    - the server SHALL support all listed values.
  - within the tables in the "Client data - Specializations" sections:
    - the client SHALL support at least one of the listed values.

Depending on the Element, different values may be used during runtime. If this is the case, those rules are described within the value rules.

If a value is placed in parenthesis, the corresponding value is a recommendation. The actual value MAY deviate from this, e.g. "(1024)".

3.1.3.8 General information on how to interpret the "Content of Function..." and "Content of Specialization..." tables

Within the "Client data - Specializations" sections each Specialization is described in an own sub-section with the name "Specialization "<name of the Specialization>"" (e.g. "Specialization "Measurement_GridFeedInEnergy""). It contains only one table that includes all Elements needed for this Specialization. The different Functions are mentioned in a continuous row, highlighted with grey background colour. This row contains the following parts:

<Feature Type>. <Function>.[<list entry instance name>.]

The <list entry instance name> is only included if the <Function> is a list-based Function. An example could be:

DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData.
deviceConfigurationKeyValueDescriptionData.

In the following rows, only the names of the Elements are stated, without the prefix described above.

Within the "Server data - Resources" sections each Feature Type is described in an own sub-section with the name "Feature Type "<name of the Feature Type>"" (e.g. "Feature Type "Measurement""). It contains sub-sections for each Function named "Function "<name of the Function>"" (e.g. "Function "measurementListData""). These sections contain one table with all Elements needed for this resource. The list entries are mentioned in a continuous row, highlighted with grey background colour. This row contains the following parts:

<Feature Type>. <Function>.[<list entry instance name>.]

The <list entry instance name> is only included if the <Function> is a list-based Function. An example could be:

Measurement. measurementDescriptionListData. measurementDescriptionData.

In the following rows, only the names of the Elements are stated, without the prefix described above.

For both kinds of tables, the following applies:

- Parent Elements are marked with a dot at the end of the name:
  <parent Element>.
  E.g.:
  value.
- If there are sub-Elements, they are described in own rows with the name of the parent Element as prefix, separated by a dot and a blank space:

<parent Element>. <sub-Element>

E.g.:

value. number

3.1.4 Rules for "Feature Types and Functions..." tables

3.1.4.1 Presence indications for "Feature Types and Functions..." tables

The following presence indications are used:

Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 14: Presence indication of Feature Types and Functions support

If at least one Function of a Feature has the presence indication "M", it is mandatory to support the Feature.

3.1.4.2 Rules for "Possible operations" column

Within the "Feature Types and Functions..." tables the column "Possible operations" state whether the Function is read- or writeable (as defined in the detailed discovery mechanism, see [ProtocolSpecification]).

If the "partial" concept (also called "restricted function exchange") SHALL be supported, the following notation is used (separated for read and write access):

read (M). partial (M)
write (M). partial (M)

If the "partial" concept SHOULD be supported, the following notation is used:

read (M). partial (R)
write (M). partial (R)

If the "partial" concept MAY be supported, the following notation is used:

read (M). partial (O)
write (M). partial (O)

The server can decide whether a notification is submitted complete or partial (as described in [ProtocolSpecification]) if not defined differently within this Use Case Specification.

3.1.5 "Actor ... overview" diagram rules

Within the "Actor [...] overview" diagrams in the "Actors" sub-sections the complete functionality of this Use Case is provided, including optional Scenarios. Which Scenarios are optional can be found in Table 1. The Actor MAY have more functionality implemented than needed for this Use Case.

For the following Actor overview example, a brief description of the graphical symbols will be described.

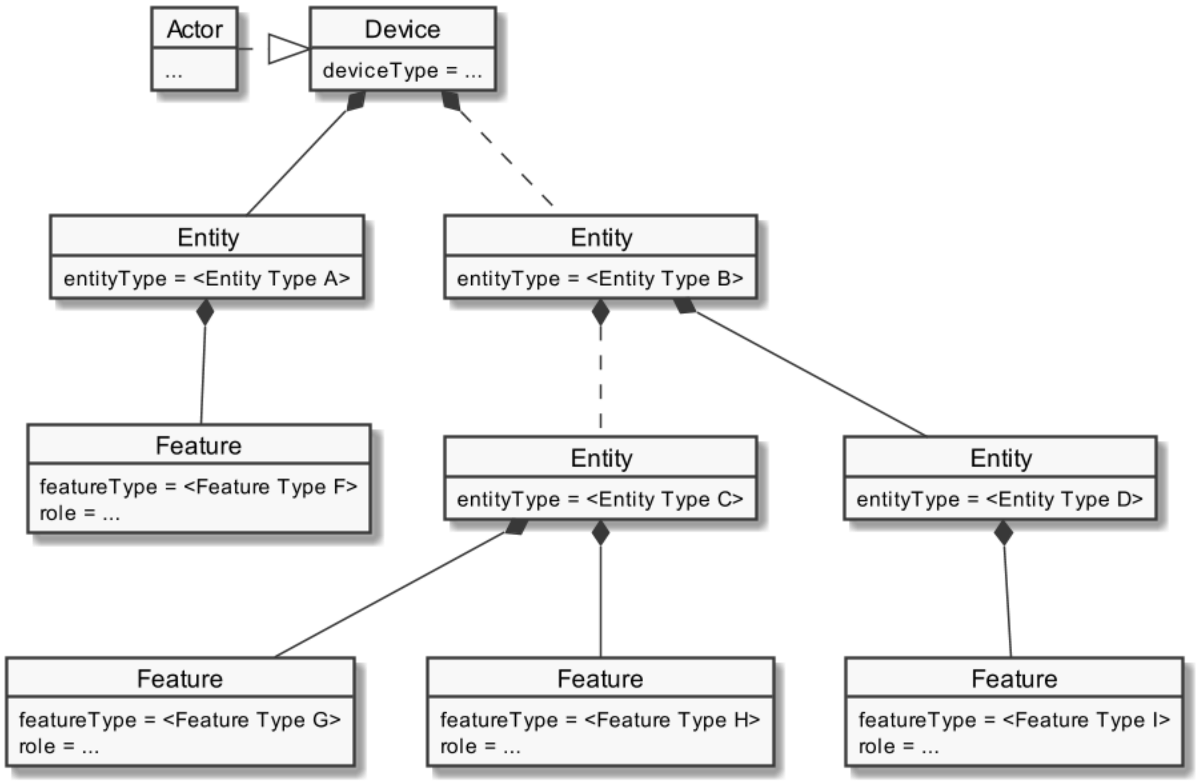


Figure 4: Actor overview example

The solid lines in the figure represent an immediate parent-childhood relation: The Entity with "<Entity Type A>" is a direct child of "Device". The Entity with "<Entity Type D>" is a direct child of the Entity with "<Entity Type B>". All Features are immediate child of the respective Entity.

The dashed lines in the figure express that there MAY be additional Entities between the shown Entities: A vendor's implementation MAY have one or more Entities between "Device" and the Entity with "<Entity Type B>". Likewise, a vendor's implementation MAY have one or more Entities between the Entity with "<Entity Type B>" and the Entity with "<Entity Type C>".

3.1.6 Specializations

Within the "Actors" sub-sections, Specializations are referenced. A Specialization describes a dataset necessary to fulfil the specific requirements of a High-Level Use Case and its Scenarios. Often, data from multiple different Features and Functions are needed to fulfil the requirements. Therefore, a Specialization defines a dataset that may encompass multiple related Functions from one or more different Features.

As different Use Cases sometimes share similar requirements, Specializations are also important from a re-usability perspective. This approach is used to improve consistency across Use Cases and avoid multiple variances of basically the same dataset. This is especially important in the case when an implementation supports multiple Use Cases. E.g. if a power measurement is necessary in two different Use Cases, both Use Cases could define slightly different datasets. In this case, the server as well as the client functionality would have to implement both variances if both Use Cases are supported. This means, depending on the number of Use Cases, two or more datasets need to be generated, transmitted and stored instead of one. Therefore, common Specializations are defined to enable consolidated implementations. Specializations are detailed per Actor, as distinct Actors may use different parts of a Specialization or even have different permissions to change data. In general, this may also depend on the respective Scenario.

If a Feature server can provide the data of a Specialization, the data does not necessarily always need to be available at the Feature server. There might be situations where the user deactivates a Use Case. There may also be other reasons why Use Case data cannot be provided currently. Therefore, a client always needs to be subscribed (as described in section 3.3.4) on the corresponding dataset to stay updated.

The SPINE resource descriptions given in the "SPINE resources of the Actor" sections are derived from the Specializations given in the Actor's overview diagram.

3.1.7 Order of messages within the sequence diagrams

There are several sequence diagrams in this document describing message flows. The order of the messages SHOULD be kept by the communications partners, but there might be cases where a different order makes sense. The communications partners SHALL be able to handle the Scenario functionalities even if the messages are transmitted in a different order by the other Actor(s). The sequence diagrams can be seen as examples.

3.1.8 Further information and rules

3.1.8.1 Frequently used Element rules for the Resource and Specialization tables

This section serves as a collection of rules frequently used by Resource and Specialization tables of the subsequent sections. Each rule applies only where referenced explicitly in the tables.

Note: The purpose of this collection is just to reduce the size of the tables. As such, no rule has a meaning without a reference indicating the required rule. A reference looks like "See [Measurement value rules]", e.g.

[Measurement value rules]:

SHALL be set if a value is available. Otherwise, the whole list entry SHALL be omitted or the Element *valueState* SHALL be set to "error".

If *valueState* is set to "error", but *value* is set, the content of *value* SHALL be ignored by the client.

If *valueState* is set to "outOfRange", but *value* is set, the content of *value* SHALL be interpreted as being out of range.

If *valueState* is set to "outOfRange", *measurementConstraintsListData. valueRangeMax* is set and *value* is equal or bigger than *valueRangeMax*, value SHALL be interpreted as above *valueRangeMax*.

If *valueState* is set to "outOfRange", *measurementConstraintsListData. valueRangeMin* is set and *value* is equal or smaller than *valueRangeMin*, value SHALL be interpreted as below *valueRangeMin*.

If set, *measurementDescriptionListData. measurementType* SHALL be set, too.

[Value state rules]:

The Element valueState SHALL be set if its content differs from "normal" ([MOB-005]). This means, if the state of the value is "outOfRange" or "error" this SHALL be denoted in the valueState Element. A client SHALL always consider the content of valueState, if set. If omitted, a value of "normal" is assumed.

[String-length rules]:

The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 256 characters.

[Scaled number rules]:

The sub-Elements "number" and "scale" represent a value according to the following formula: number * 10^{scale}

[Evaluation period rules]:

The "evaluationPeriod" expresses the analysis period for the according value.

3.1.8.2 Rules regarding the usage of time-related information

All time-related information within this Use Case SHALL be presented as absolute times. This means relative times SHALL NOT be used, unless the data point only supports relative times (e.g. for a "duration").

Note: This means the Actors need to have a common understanding of these absolute times. It is strongly recommended that each Actor is "sufficiently synchronised" with the official "Coordinated Universal Time" (UTC).

3.1.8.3 Applied sign convention

For the applied sign convention, please see section 2.6.1.

3.2 Actors

3.2.1 Monitoring Appliance

3.2.1.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "MonitoringAppliance" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Monitoring Appliance's resource hierarchy.

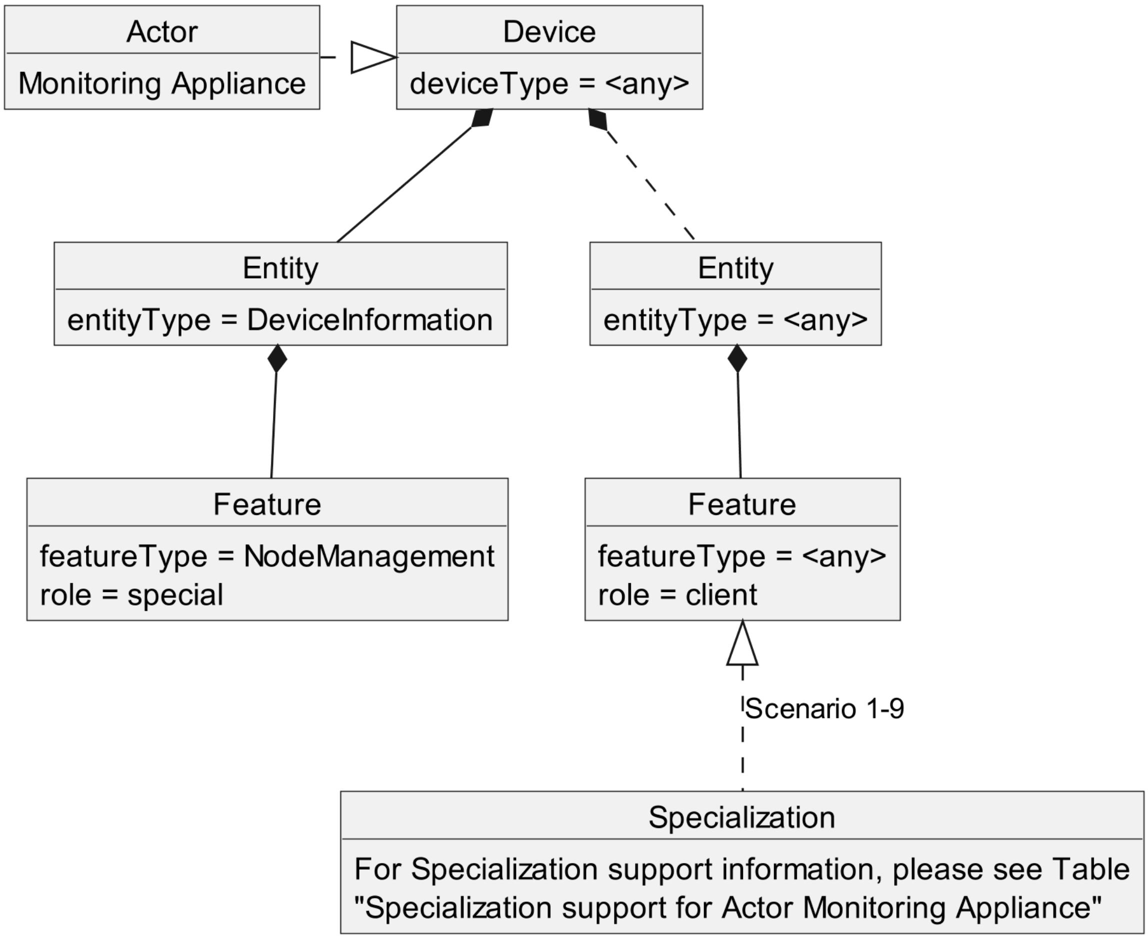


Figure 5: Actor "Monitoring Appliance" overview

The ""Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follows behind any entityType. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	**Scenario**	**Described in table**
DeviceClassification_ManufacturerData_MOB	1	Table 16
DeviceDiagnosis_OperatingState	2	Table 17
Measurement_StateOfCharge	2	Table 18
Measurement_StateOfHealth	2	Table 19
Measurement_StateOfEnergy	2	Table 20
Measurement_UseableCapacity	2	Table 21
Measurement_DcPower	3	Table 22
Measurement_DcCurrent	4	Table 23
Measurement_DcVoltage	5	Table 24
Measurement_DcChargeEnergy	6	Table 25
Measurement_DcDischargeEnergy	6	Table 26
Measurement_LoadCycleCount	7	Table 27
Measurement_ComponentTemperature	9	Table 28
DeviceConfiguration_BatteryType	7	Table 29
DeviceConfiguration_MaxCycleCountPerDay	8	Table 30
ElectricalConnection_EntityEnergyCapacityNominalMax	8	Table 31
ElectricalConnection_EntityPowerConsumptionNominalMax	8	Table 32
ElectricalConnection_EntityPowerProductionNominalMax	8	Table 33

Table 15: Specialization support for Actor Monitoring Appliance

3.2.1.2 Server data - Resources

As this Actor has only client functionality, no resources are described within this section.

3.2.1.3 Client data - Specializations

3.2.1.3.1 Topic "DeviceClassification"

3.2.1.3.1.1 Specialization "DeviceClassification_ManufacturerData_MOB"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M	DeviceClassification. deviceClassificationManufacturerData.		
1: R	deviceName		[MOB-011]

			[String-length rules]
1: R	deviceCode		[MOB-012] [String-length rules]
1: R	serialNumber		[MOB-013] [String-length rules]
1: R	softwareRevision		[MOB-014] [String-length rules]
1: R	hardwareRevision		[MOB-015] [String-length rules]
1: M	vendorName		[MOB-016] [String-length rules]
1: O	vendorCode		[String-length rules]
1: O	brandName		[String-length rules]
1: O	manufacturerLabel		[MOB-017] [String-length rules]
1: O	manufacturerDescription		The string-length SHOULD NOT be longer than 4096 characters.

Table 16: Content of Specialization "DeviceClassification_ManufacturerData_MOB" at Actor Monitoring Appliance

3.2.1.3.2 Topic "DeviceDiagnosis"

3.2.1.3.2.1 Specialization "DeviceDiagnosis_OperatingState"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	DeviceDiagnosis. deviceDiagnosisStateData.		
2: O	timestamp		
2: M	operatingState	"normalOperation" [MOB-025/1]	[MOB-025]
		"standby" [MOB-025/2]	
		"failure" [MOB-025/5]	
		"temporarilyNotReady" [MOB-025/3]	
		"off" [MOB-025/4]	
2: O	vendorStateCode		The string-length SHOULD NOT be longer than 128 characters.

Table 17: Content of Specialization "DeviceDiagnosis_OperatingState" at Actor Monitoring Appliance

3.2.1.3.3 Topic "Measurement"

3.2.1.3.3.1 Specialization "Measurement_StateOfCharge"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"percentage"	
2: M	commodityType	"electricity"	
2: M	unit	"pct"	
2: M	scopeType	"stateOfCharge"	
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: R	valueRangeMin.		[MOB-002/2] SHOULD be used. [Scaled number rules]
2: M	valueRangeMin. number		SHALL be used.
2: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueRangeMax.		[MOB-002/2] SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.
2: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.
2: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	Measurement. measurementListData. measurementData.		
2: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#(1..1)->m1#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
2: M	value.		[MOB-002/2], [MOB-021] [Measurement value rules] [Scaled number rules]
2: M	value. number		SHALL be used.

2: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueSource	"calculatedValue"	
2: M	valueState		[MOB-005] [Value state rules]

Table 18: Content of Specialization "Measurement_StateOfCharge" at Actor Monitoring Appliance

3.2.1.3.3.2 Specialization "Measurement_StateOfHealth"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"percentage"	
2: M	unit	"pct"	
2: M	scopeType	"stateOfHealth"	
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: R	valueRangeMin.		[MOB-002/2] SHOULD be used. [Scaled number rules]
2: M	valueRangeMin. number		SHALL be used.
2: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueRangeMax.		[MOB-002/2] SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.
2: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.
2: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	Measurement. measurementListData. measurementData.		
2: M	measurementId	<m2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#(1..1)->m2#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
2: M	value.		[MOB-002/2], [MOB-022]

			[Measurement value rules] [Scaled number rules]
2: M	value. number		SHALL be used.
2: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
2: M	valueState		[MOB-005] [Value state rules]

Table 19: Content of Specialization "Measurement_StateOfHealth" at Actor Monitoring Appliance

3.2.1.3.3.3 Specialization "Measurement_StateOfEnergy"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"energy"	
2: M	commodityType	"electricity"	
2: M	unit	"Wh"	
2: M	scopeType	"stateOfEnergy"	
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: R	valueRangeMin.		[MOB-002/3] SHOULD be used. [Scaled number rules]
2: M	valueRangeMin. number		SHALL be used.
2: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueRangeMax.		[MOB-002/3] SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.
2: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.
2: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	Measurement. measurementListData. measurementData.		
2: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.

2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#(1..1)->m3#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
2: M	value.		[MOB-002/3], [MOB-023] [Measurement value rules] [Scaled number rules]
2: M	value. number		SHALL be used.
2: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
2: M	valueState		[MOB-005] [Value state rules]

Table 20: Content of Specialization "Measurement_StateOfEnergy" at Actor Monitoring Appliance

3.2.1.3.3.4 Specialization "Measurement_UseableCapacity"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"capacity"	
2: M	commodityType	"electricity"	
2: M	unit	"Wh"	
2: M	scopeType	"useableCapacity"	
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: R	valueRangeMin.		[MOB-002/3] SHOULD be used. [Scaled number rules]
2: M	valueRangeMin. number		SHALL be used.
2: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueRangeMax.		[MOB-002/3] SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.
2: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.

2: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	Measurement. measurementListData. measurementData.		
2: M	measurementId	<m4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#(1..1)->m4#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
2: M	value.		[MOB-002/3], [MOB-024] [Measurement value rules] [Scaled number rules]
2: M	value. number		SHALL be used.
2: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
2: M	valueState		[MOB-005] [Value state rules]
2: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
2: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	powerSupplyType	"dc"	
2: M	positiveEnergyDirection	"consume"	
2: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
2: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p1#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
2: M	measurementId	<m4#(1..1)->p1#1>	SHALL be set as FOREIGN IDENTIFIER.

Table 21: Content of Specialization "Measurement_UseableCapacity" at Actor Monitoring Appliance

3.2.1.3.3.5 Specialization "Measurement_DcPower"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"power"	
3: M	commodityType	"electricity"	
3: M	unit	"W"	

3: M	scopeType	"dcPower"	
3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: R	valueRangeMin.		[MOB-001] SHOULD be used. [Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueRangeMax.		[MOB-001] SHOULD be used. [Scaled number rules]
3: M	valueRangeMax. number		SHALL be used.
3: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.
3: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#(1..1)->m5#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MOB-001], [MOB-031] [Measurement value rules] [Scaled number rules]
3: M	value. number		SHALL be used.
3: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
3: M	valueState		[MOB-005] [Value state rules]
3: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
3: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	powerSupplyType	"dc"	
3: M	positiveEnergyDirection	"consume"	
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p2#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.

3: M	measurementId	<m5#(1..1)->p2#1>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"dc"	

Table 22: Content of Specialization "Measurement_DcPower" at Actor Monitoring Appliance

3.2.1.3.3.6 Specialization "Measurement_DcCurrent"

Scenario [{…}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"current"	
4: M	commodityType	"electricity"	
4: M	unit	"A"	
4: M	scopeType	"dcCurrent"	
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MOB-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MOB-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#(1..1)->m6#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
4: M	value.		[MOB-001], [MOB-041] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.

4: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: M	valueState		[MOB-005] [Value state rules]
4: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	powerSupplyType	"dc"	
4: M	positiveEnergyDirection	"consume"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p3#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m6#(1..1)->p3#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"dc"	

Table 23: Content of Specialization "Measurement_DcCurrent" at Actor Monitoring Appliance

3.2.1.3.3.7 Specialization "Measurement_DcVoltage"

Scenario [{…}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
5: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
5: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	measurementType	"voltage"	
5: M	commodityType	"electricity"	
5: M	unit	"V"	
5: M	scopeType	"dcVoltage"	
5: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
5: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: R	valueRangeMin.		[MOB-002/1] SHOULD be used. [Scaled number rules]
5: M	valueRangeMin. number		SHALL be used.
5: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: R	valueRangeMax.		[MOB-002/1] SHOULD be used. [Scaled number rules]
5: M	valueRangeMax. number		SHALL be used.
5: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: R	valueStepSize.		SHOULD be used. [Scaled number rules]

5: M	valueStepSize. number		SHALL be used.
5: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: M	Measurement. measurementListData. measurementData.		
5: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
5: O	timestamp	<t#(1..1)->m7#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
5: M	value.		[MOB-002/1], [MOB-051] [Measurement value rules] [Scaled number rules]
5: M	value. number		SHALL be used.
5: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
5: M	valueState		[MOB-005] [Value state rules]
5: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
5: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	powerSupplyType	"dc"	
5: M	positiveEnergyDirection	"consume"	
5: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
5: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	parameterId	<p4#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
5: M	measurementId	<m7#(1..1)->p4#1>	SHALL be set as FOREIGN IDENTIFIER.
5: M	voltageType	"dc"	

Table 24: Content of Specialization "Measurement_DcVoltage" at Actor Monitoring Appliance

3.2.1.3.3.8 Specialization "Measurement_DcChargeEnergy"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
6: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
6: M	measurementId	<m8#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	measurementType	"energy"	
6: M	commodityType	"electricity"	
6: M	unit	"Wh"	
6: M	scopeType	"dcChargeEnergy"	
6: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
6: M	measurementId	<m8#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.

6: R	valueRangeMin.		[MOB-001] SHOULD be used. [Scaled number rules]
6: M	valueRangeMin. number		SHALL be used.
6: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: R	valueRangeMax.		[MOB-001] SHOULD be used. [Scaled number rules]
6: M	valueRangeMax. number		SHALL be used.
6: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: R	valueStepSize.		SHOULD be used. [Scaled number rules]
6: M	valueStepSize. number		SHALL be used.
6: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: M	Measurement. measurementListData. measurementData.		
6: M	measurementId	<m8#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
6: O	timestamp	<t#(1..1)->m8#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
6: M	value.		[MOB-001], [MOB-061] [Measurement value rules] [Scaled number rules]
6: M	value. number		SHALL be used.
6: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: O	evaluationPeriod.		[Evaluation period rules]
6: M	evaluationPeriod. startTime		
6: M	evaluationPeriod. endTime		
6: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
6: M	valueState		[MOB-005] [Value state rules]
6: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	powerSupplyType	"dc"	
6: M	positiveEnergyDirection	"consume"	
6: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p5#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
6: M	measurementId	<m8#(1..1)->p5#1>	SHALL be set as FOREIGN IDENTIFIER.

6: M	voltageType	"dc"	

Table 25: Content of Specialization "Measurement_DcChargeEnergy" at Actor Monitoring Appliance

3.2.1.3.3.9 Specialization "Measurement_DcDischargeEnergy"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
6: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
6: M	measurementId	<m9#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	measurementType	"energy"	
6: M	commodityType	"electricity"	
6: M	unit	"Wh"	
6: M	scopeType	"dcDischargeEnergy"	
6: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
6: M	measurementId	<m9#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: R	valueRangeMin.		[MOB-001] SHOULD be used. [Scaled number rules]
6: M	valueRangeMin. number		SHALL be used.
6: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: R	valueRangeMax.		[MOB-001] SHOULD be used. [Scaled number rules]
6: M	valueRangeMax. number		SHALL be used.
6: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: R	valueStepSize.		SHOULD be used. [Scaled number rules]
6: M	valueStepSize. number		SHALL be used.
6: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: M	Measurement. measurementListData. measurementData.		
6: M	measurementId	<m9#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
6: O	timestamp	<t#(1..1)->m9#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
6: M	value.		[MOB-001], [MOB-062] [Measurement value rules] [Scaled number rules]
6: M	value. number		SHALL be used.

6: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
6: O	evaluationPeriod.		[Evaluation period rules]
6: M	evaluationPeriod. startTime		
6: M	evaluationPeriod. endTime		
6: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
6: M	valueState		[MOB-005] [Value state rules]
6: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	powerSupplyType	"dc"	
6: M	positiveEnergyDirection	"consume"	
6: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p6#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
6: M	measurementId	<m9#(1..1)->p6#1>	SHALL be set as FOREIGN IDENTIFIER.
6: M	voltageType	"dc"	

1002 *Table 26: Content of Specialization "Measurement_DcDischargeEnergy" at Actor Monitoring Appliance*

1003

1004 *3.2.1.3.3.10 Specialization "Measurement_LoadCycleCount"*

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
7: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
7: M	measurementId	<m10#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
7: M	measurementType	"count"	
7: M	unit	"1"	
7: M	scopeType	"loadCycleCount"	
7: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
7: M	measurementId	<m10#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
7: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
7: M	valueRangeMin. number	≥0	SHALL be used. The value of Element "number" SHALL be greater than or equal to zero.
7: M	valueRangeMin. scale	≥0	SHALL be interpreted. If absent, a default value of "0" applies. If present, the value of

			Element "scale" SHALL be greater than or equal to zero.
7: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
7: M	valueRangeMax. number	≥0	SHALL be used. The value of Element "number" SHALL be greater than or equal to zero.
7: M	valueRangeMax. scale	≥0	SHALL be interpreted. If absent, a default value of "0" applies. If present, the value of Element "scale" SHALL be greater than or equal to zero.
7: R	valueStepSize.		SHOULD be used. [Scaled number rules]
7: M	valueStepSize. number	≥0	SHALL be used. The value of Element "number" SHALL be greater than or equal to zero.
7: M	valueStepSize. scale	≥0	SHALL be interpreted. If absent, a default value of "0" applies. If present, the value of Element "scale" SHALL be greater than or equal to zero.
7: M	Measurement. measurementListData. measurementData.		
7: M	measurementId	<m10#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
7: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
7: O	timestamp	<t#(1..1)->m10#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
7: M	value.		[MOB-071] [Measurement value rules] [Scaled number rules]
7: M	value. number	≥0	SHALL be used. The value of Element "number" SHALL be greater than or equal to zero.
7: M	value. scale	≥0	SHALL be interpreted. If absent, a default value of "0" applies. If present, the value of Element "scale" SHALL be greater than or equal to zero.
7: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
7: M	valueState		[MOB-005] [Value state rules]

1005 *Table 27: Content of Specialization "Measurement_LoadCycleCount" at Actor Monitoring Appliance*

1006

1007 *3.2.1.3.3.11 Specialization "Measurement_ComponentTemperature"*

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**

9: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
9: M	measurementId	<m11#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
9: M	measurementType	"temperature"	
9: M	unit	"degC" | "degF" | "K"	
9: M	scopeType	"componentTemperature"	
9: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
9: M	measurementId	<m11#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
9: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
9: M	valueRangeMin. number		SHALL be used.
9: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
9: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
9: M	valueRangeMax. number		SHALL be used.
9: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
9: R	valueStepSize.		SHOULD be used. [Scaled number rules]
9: M	valueStepSize. number		SHALL be used.
9: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
9: M	Measurement. measurementListData. measurementData.		
9: M	measurementId	<m11#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
9: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
9: O	timestamp	<t#(1..1)->m11#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
9: M	value.		[MOB-091] [Measurement value rules] [Scaled number rules]
9: M	value. number		SHALL be used.
9: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
9: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
9: M	valueState		[MOB-005] [Value state rules]

Table 28: Content of Specialization "Measurement_ComponentTemperature" at Actor Monitoring Appliance

3.2.1.3.4 Topic "DeviceConfiguration"

3.2.1.3.4.1 Specialization "DeviceConfiguration_BatteryType"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
7: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
7: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
7: M	keyName	"batteryType"	
7: M	valueType	"string"	
7: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
7: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
7: M	value.		[MOB-072] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
7: M	value. string	"liIon"	[MOB-072/1]
		"leadAcid"	[MOB-072/2]
		"redoxFlow"	[MOB-072/3]
		"sodiumIon"	[MOB-072/4]

Table 29: Content of Specialization "DeviceConfiguration_BatteryType" at Actor Monitoring Appliance

3.2.1.3.4.2 Specialization "DeviceConfiguration_MaxCycleCountPerDay"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
8: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
8: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
8: M	keyName	"maxCycleCountPerDay"	
8: M	valueType	"scaledNumber"	
8: M	unit	"1"	The unit SHALL be applied to the value of the key.
8: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
8: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
8: M	value.		[MOB-084] Exactly one of the child Elements SHALL be set. This SHALL match with the content of

			valueType Element within the key value description part (see above).
8: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
8: M	value. scaledNumber. number	≥0	SHALL be used. The value of Element "number" SHALL be greater than or equal to zero.
8: M	value. scaledNumber. scale	≥0	SHALL be interpreted. If absent, a default value of "0" applies. If present, the value of Element "scale" SHALL be greater than or equal to zero.

Table 30: Content of Specialization "DeviceConfiguration_MaxCycleCountPerDay" at Actor Monitoring Appliance

3.2.1.3.5 Topic "ElectricalConnection"

3.2.1.3.5.1 Specialization "ElectricalConnection_EntityEnergyCapacityNominalMax"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
8: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
8: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
8: M	parameterId	<p1#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
8: M	characteristicId	<cc1#(1..1)->p1#1>	SHALL be set as SUB IDENTIFIER.
8: M	characteristicContext	"entity"	
8: M	characteristicType	"energyCapacityNominalMax"	
8: M	value.		[MOB-002/3], [MOB-081] [Scaled number rules]
8: M	value. number		SHALL be used.
8: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
8: M	unit	"Wh"	

Table 31: Content of Specialization "ElectricalConnection_EntityEnergyCapacityNominalMax" at Actor Monitoring Appliance

3.2.1.3.5.2 Specialization "ElectricalConnection_EntityPowerConsumptionNominalMax"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
8: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
8: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
8: M	parameterId	<p2#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
8: M	characteristicId	<cc2#(1..1)->p2#1>	SHALL be set as SUB IDENTIFIER.
8: M	characteristicContext	"entity"	
8: M	characteristicType	"powerConsumptionNominalMax"	
8: M	value.		[MOB-003], [MOB-082] [Scaled number rules]
8: M	value. number		SHALL be used.
8: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
8: M	unit	"W"	

Table 32: Content of Specialization "ElectricalConnection_EntityPowerConsumptionNominalMax" at Actor Monitoring Appliance

3.2.1.3.5.3 Specialization "ElectricalConnection_EntityPowerProductionNominalMax"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
8: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
8: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
8: M	parameterId	<p2#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
8: M	characteristicId	<cc3#(1..1)->p2#1>	SHALL be set as SUB IDENTIFIER.
8: M	characteristicContext	"entity"	
8: M	characteristicType	"powerProductionNominalMax"	
8: M	value.		[MOB-003], [MOB-083] [Scaled number rules]
8: M	value. number		SHALL be used.

8: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
8: M	unit	"W"	

Table 33: Content of Specialization "ElectricalConnection_EntityPowerProductionNominalMax" at Actor Monitoring Appliance

3.2.2 Battery

3.2.2.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "Battery" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Battery's resource hierarchy.

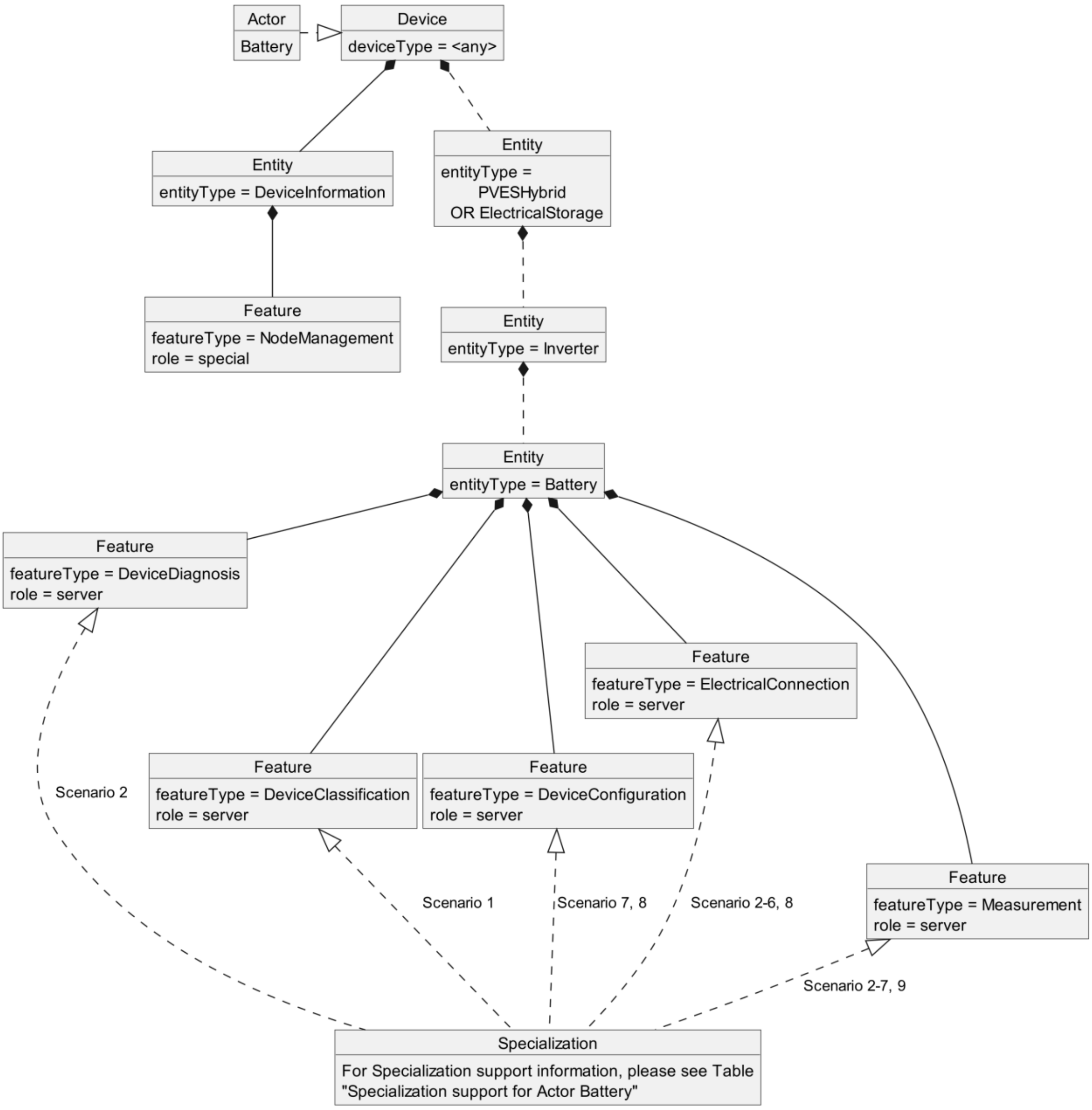


Figure 6: Actor "Battery" overview

The ""Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follows behind the entityType "Battery", which is a sub-Entity of the Entity "Inverter" that is a sub-Entity of one of the following Entities: "PVESHybrid" ([MOB-004/1]) or "ElectricalStorage" ([MOB-004/2]) (see section 2.2). The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

Note: As shown in Figure 2, an Inverter may have more than one Battery attached. This means the Entity "Inverter" in Figure 6 may contain more than one Entity with entityType "Battery". In this Use Case, each "Battery" Entity represents exactly one individual Actor Battery.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	**Scenario**	**Used Feature...**	**...in tables**
DeviceClassification_ManufacturerData_MOB	1	DeviceClassification	Table 36
DeviceDiagnosis_OperatingState	2	DeviceDiagnosis	Table 39
Measurement_StateOfCharge	2	Measurement	Table 43 Table 44 Table 45
Measurement_StateOfHealth	2		
Measurement_StateOfEnergy	2		
Measurement_UseableCapacity	2	ElectricalConnection Measurement	Table 40 Table 41 Table 43 Table 44 Table 45
Measurement_DcPower	3		
Measurement_DcCurrent	4		
Measurement_DcVoltage	5		
Measurement_DcChargeEnergy	6		
Measurement_DcDischargeEnergy	6		
Measurement_LoadCycleCount	7	Measurement	Table 43 Table 44 Table 45
Measurement_ComponentTemperature	9		
DeviceConfiguration_BatteryType	7	DeviceConfiguration	Table 37 Table 38
DeviceConfiguration_MaxCycleCountPerDay	8		
ElectricalConnection_EntityEnergyCapacityNominalMax	8	ElectricalConnection	Table 42
ElectricalConnection_EntityPowerConsumptionNominalMax	8		

ElectricalConnection_EntityPowerProductionNominalMax	8		

Table 34: Specialization support for Actor Battery

3.2.2.2 Server data - Resources

3.2.2.2.1 Overview

Behind the entityType "Battery", the Actor Battery SHALL offer the Feature Types and Functions given in the table below.

Feature Type	**Scenario: M/R/O**	**Function**	**Possible operations**
DeviceClassification	1: M	deviceClassificationManufacturerData	read (M)
DeviceConfiguration	7: M 8: M	deviceConfigurationKeyValueDescriptionListData	read (M). partial (R)
	7: M 8: M	deviceConfigurationKeyValueListData	read (M). partial (R)
DeviceDiagnosis	2: M	deviceDiagnosisStateData	read (M)
ElectricalConnection	2: M 3: M 4: M 5: M 6: M	electricalConnectionDescriptionListData	read (M). partial (R)
	2: M 3: M 4: M 5: M 6: M	electricalConnectionParameterDescriptionListData	read (M). partial (R)
	8: M	electricalConnectionCharacteristicListData	read (M). partial (R)
Measurement	2: M 3: M 4: M 5: M 6: M 7: M 9: M	measurementDescriptionListData	read (M). partial (R)
	2: R 3: R 4: R 5: R 6: R 7: R 9: R	measurementConstraintsListData	read (M). partial (R)
	2: M 3: M 4: M 5: M 6: M 7: M 9: M	measurementListData	read (M). partial (R)

Table 35: Feature Types and Functions used within this Use Case by the Actor Battery

For each of these Feature Types, the following rule applies: There SHALL be at maximum one Feature with the Feature Type in the Entity.

Note: As a consequence of the previous rule, an implementation may need to have Feature data from different Scenarios/Specializations or even Use Cases in a given Feature.

The Scenario number shows in which Scenarios the Battery acts as server and which Feature Types and Functions are relevant in each Scenario.

A detailed definition of the Elements and values that shall be supported in each Function is given in the following sub-sections.

Note: If in the table above "partial" read is not mentioned or is only optional, it still might be mandatory to support partial notifications. The details of "partial" support are described within the Scenario sections.

Note: The presence indications stated above are meant relative to the ones of the according Scenario stated in Table 1. I.e., if a Scenario is optional ("O") and a Feature Type is mandatory ("M"), the Feature Type need only be supported if the Scenario is supported, too.

Note: Further Features MAY be implemented on the same Entities; also, further Functions MAY be implemented in the used Entities.

3.2.2.2.2 Feature Type "DeviceClassification"

3.2.2.2.2.1 Function "deviceClassificationManufacturerData"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
1: M	DeviceClassification. deviceClassificationManufacturerData.		
1: R	deviceName		[MOB-011] [String-length rules]
1: R	deviceCode		[MOB-012] [String-length rules]
1: R	serialNumber		[MOB-013] [String-length rules]
1: R	softwareRevision		[MOB-014] [String-length rules]
1: R	hardwareRevision		[MOB-015] [String-length rules]
1: M	vendorName		[MOB-016] [String-length rules]
1: O	vendorCode		[String-length rules]
1: O	brandName		[String-length rules]

1: O	manufacturerLabel		[MOB-017] [String-length rules]
1: O	manufacturerDescription		The string-length SHOULD NOT be longer than 4096 characters.

Table 36: Content of Function "deviceClassificationManufacturerData" at Actor Battery

3.2.2.2.3 Feature Type "DeviceConfiguration"

3.2.2.2.3.1 Function "deviceConfigurationKeyValueDescriptionListData"

Scenario [{…}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
7: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
7: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
7: M	keyName	"batteryType"	
7: M	valueType	"string"	
8: M	DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData.		
8: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
8: M	keyName	"maxCycleCountPerDay"	
8: M	valueType	"scaledNumber"	
8: M	unit	"1"	The unit SHALL be applied to the value of the key.

Table 37: Content of Function "deviceConfigurationKeyValueDescriptionListData" at Actor Battery

3.2.2.2.3.2 Function "deviceConfigurationKeyValueListData"

Scenario [{…}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
7: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
7: M	keyId	<k1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
7: M	value.		[MOB-072] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
7: M	value. string	"liIon"	[MOB-072/1]
		"leadAcid"	[MOB-072/2]
		"redoxFlow"	[MOB-072/3]
		"sodiumIon"	[MOB-072/4]

8: M	DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData.		
8: M	keyId	<k2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
8: M	value.		[MOB-084] Exactly one of the child Elements SHALL be set. This SHALL match with the content of *valueType* Element within the key value description part (see above).
8: M	value. scaledNumber.		SHALL be used. [Scaled number rules]
8: M	value. scaledNumber. number	≥0	SHALL be used. The value of Element "number" SHALL be greater than or equal to zero.
8: O	value. scaledNumber. scale	≥0	MAY be used. If absent, a default value of "0" applies. If present, the value of Element "scale" SHALL be greater than or equal to zero.

Table 38: Content of Function "deviceConfigurationKeyValueListData" at Actor Battery

3.2.2.2.4 Feature Type "DeviceDiagnosis"

3.2.2.2.4.1 Function "deviceDiagnosisStateData"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	DeviceDiagnosis. deviceDiagnosisStateData.		
2: O	timestamp		
2: M	operatingState	"normalOperation" [MOB-025/1]	[MOB-025]
		"standby" [MOB-025/2]	
		"failure" [MOB-025/5]	
		"temporarilyNotReady" [MOB-025/3]	
		"off" [MOB-025/4]	
2: O	vendorStateCode		The string-length SHOULD NOT be longer than 128 characters.

Table 39: Content of Function "deviceDiagnosisStateData" at Actor Battery

3.2.2.2.5 Feature Type "ElectricalConnection"

3.2.2.2.5.1 Function "electricalConnectionDescriptionListData"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**

2: M 3: M 4: M 5: M 6: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
2: M 3: M 4: M 5: M 6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M 3: M 4: M 5: M 6: M	powerSupplyType	"dc"	
2: M 3: M 4: M 5: M 6: M	positiveEnergyDirection	"consume"	

Table 40: Content of Function "electricalConnectionDescriptionListData" at Actor Battery

3.2.2.2.5.2 Function "electricalConnectionParameterDescriptionListData"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
2: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p1#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
2: M	measurementId	<m4#(1..1)->p1#1>	SHALL be set as FOREIGN IDENTIFIER.
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p2#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
3: M	measurementId	<m5#(1..1)->p2#1>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"dc"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.

4: M	parameterId	<p3#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m6#(1..1)->p3#1>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"dc"	
5: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
5: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	parameterId	<p4#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
5: M	measurementId	<m7#(1..1)->p4#1>	SHALL be set as FOREIGN IDENTIFIER.
5: M	voltageType	"dc"	
6: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p5#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
6: M	measurementId	<m8#(1..1)->p5#1>	SHALL be set as FOREIGN IDENTIFIER.
6: M	voltageType	"dc"	
6: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
6: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	parameterId	<p6#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
6: M	measurementId	<m9#(1..1)->p6#1>	SHALL be set as FOREIGN IDENTIFIER.
6: M	voltageType	"dc"	

Table 41: Content of Function "electricalConnectionParameterDescriptionListData" at Actor Battery

3.2.2.2.5.3 Function "electricalConnectionCharacteristicListData"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
8: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
8: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
8: M	parameterId	<p1#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
8: M	characteristicId	<cc1#(1..1)->p1#1>	SHALL be set as SUB IDENTIFIER.

8: M	characteristicContext	"entity"	
8: M	characteristicType	"energyCapacityNominalMax"	
8: M	value.		[MOB-002/3], [MOB-081] [Scaled number rules]
8: M	value. number		SHALL be used.
8: O	value. scale		MAY be used. If absent, a default value of "0" applies.
8: M	unit	"Wh"	
8: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
8: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
8: M	parameterId	<p2#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
8: M	characteristicId	<cc2#(1..1)->p2#1>	SHALL be set as SUB IDENTIFIER.
8: M	characteristicContext	"entity"	
8: M	characteristicType	"powerConsumptionNominalMax"	
8: M	value.		[MOB-003], [MOB-082] [Scaled number rules]
8: M	value. number		SHALL be used.
8: O	value. scale		MAY be used. If absent, a default value of "0" applies.
8: M	unit	"W"	
8: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
8: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
8: M	parameterId	<p2#(1..1)->ec1#1>	SHALL be set as SUB IDENTIFIER.
8: M	characteristicId	<cc3#(1..1)->p2#1>	SHALL be set as SUB IDENTIFIER.
8: M	characteristicContext	"entity"	
8: M	characteristicType	"powerProductionNominalMax"	
8: M	value.		[MOB-003], [MOB-083] [Scaled number rules]
8: M	value. number		SHALL be used.
8: O	value. scale		MAY be used. If absent, a default value of "0" applies.
8: M	unit	"W"	

Table 42: Content of Function "electricalConnectionCharacteristicListData" at Actor Battery

3.2.2.2.6 Feature Type "Measurement"

3.2.2.2.6.1 Function "measurementDescriptionListData"

Scenario [{…}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"percentage"	
2: M	commodityType	"electricity"	
2: M	unit	"pct"	
2: M	scopeType	"stateOfCharge"	
2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"percentage"	
2: M	unit	"pct"	
2: M	scopeType	"stateOfHealth"	
2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"energy"	
2: M	commodityType	"electricity"	
2: M	unit	"Wh"	
2: M	scopeType	"stateOfEnergy"	
2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"capacity"	
2: M	commodityType	"electricity"	
2: M	unit	"Wh"	
2: M	scopeType	"useableCapacity"	
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"power"	
3: M	commodityType	"electricity"	
3: M	unit	"W"	
3: M	scopeType	"dcPower"	
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"current"	
4: M	commodityType	"electricity"	
4: M	unit	"A"	
4: M	scopeType	"dcCurrent"	
5: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
5: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	measurementType	"voltage"	
5: M	commodityType	"electricity"	
5: M	unit	"V"	
5: M	scopeType	"dcVoltage"	

6: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
6: M	measurementId	<m8#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	measurementType	"energy"	
6: M	commodityType	"electricity"	
6: M	unit	"Wh"	
6: M	scopeType	"dcChargeEnergy"	
6: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
6: M	measurementId	<m9#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	measurementType	"energy"	
6: M	commodityType	"electricity"	
6: M	unit	"Wh"	
6: M	scopeType	"dcDischargeEnergy"	
7: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
7: M	measurementId	<m10#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
7: M	measurementType	"count"	
7: M	unit	"1"	
7: M	scopeType	"loadCycleCount"	
9: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
9: M	measurementId	<m11#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
9: M	measurementType	"temperature"	
9: M	unit	"degC" | "degF" | "K"	
9: M	scopeType	"componentTemperature"	

Table 43: Content of Function "measurementDescriptionListData" at Actor Battery

3.2.2.2.6.2 Function "measurementConstraintsListData"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: R	valueRangeMin.		[MOB-002/2] SHOULD be used. [Scaled number rules]
2: M	valueRangeMin. number		SHALL be used.
2: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueRangeMax.		[MOB-002/2] SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.
2: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.

2: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: R	valueRangeMin.		[MOB-002/2] SHOULD be used. [Scaled number rules]
2: M	valueRangeMin. number		SHALL be used.
2: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueRangeMax.		[MOB-002/2] SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.
2: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.
2: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: R	valueRangeMin.		[MOB-002/3] SHOULD be used. [Scaled number rules]
2: M	valueRangeMin. number		SHALL be used.
2: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueRangeMax.		[MOB-002/3] SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.
2: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.
2: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: R	valueRangeMin.		[MOB-002/3] SHOULD be used. [Scaled number rules]
2: M	valueRangeMin. number		SHALL be used.
2: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueRangeMax.		[MOB-002/3] SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.

2: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.
2: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: R	valueRangeMin.		[MOB-001] SHOULD be used. [Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
3: R	valueRangeMax.		[MOB-001] SHOULD be used. [Scaled number rules]
3: M	valueRangeMax. number		SHALL be used.
3: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.
3: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MOB-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MOB-001] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
5: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
5: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: R	valueRangeMin.		[MOB-002/1] SHOULD be used. [Scaled number rules]
5: M	valueRangeMin. number		SHALL be used.

5: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
5: R	valueRangeMax.		[MOB-002/1] SHOULD be used. [Scaled number rules]
5: M	valueRangeMax. number		SHALL be used.
5: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
5: R	valueStepSize.		SHOULD be used. [Scaled number rules]
5: M	valueStepSize. number		SHALL be used.
5: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
6: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
6: M	measurementId	<m8#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: R	valueRangeMin.		[MOB-001] SHOULD be used. [Scaled number rules]
6: M	valueRangeMin. number		SHALL be used.
6: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
6: R	valueRangeMax.		[MOB-001] SHOULD be used. [Scaled number rules]
6: M	valueRangeMax. number		SHALL be used.
6: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
6: R	valueStepSize.		SHOULD be used. [Scaled number rules]
6: M	valueStepSize. number		SHALL be used.
6: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
6: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
6: M	measurementId	<m9#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: R	valueRangeMin.		[MOB-001] SHOULD be used. [Scaled number rules]
6: M	valueRangeMin. number		SHALL be used.
6: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
6: R	valueRangeMax.		[MOB-001] SHOULD be used. [Scaled number rules]
6: M	valueRangeMax. number		SHALL be used.
6: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
6: R	valueStepSize.		SHOULD be used. [Scaled number rules]
6: M	valueStepSize. number		SHALL be used.
6: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
7: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		

7: M	measurementId	<m10#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
7: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
7: M	valueRangeMin. number	≥0	SHALL be used. The value of Element "number" SHALL be greater than or equal to zero.
7: O	valueRangeMin. scale	≥0	MAY be used. If absent, a default value of "0" applies. If present, the value of Element "scale" SHALL be greater than or equal to zero.
7: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
7: M	valueRangeMax. number	≥0	SHALL be used. The value of Element "number" SHALL be greater than or equal to zero.
7: O	valueRangeMax. scale	≥0	MAY be used. If absent, a default value of "0" applies. If present, the value of Element "scale" SHALL be greater than or equal to zero.
7: R	valueStepSize.		SHOULD be used. [Scaled number rules]
7: M	valueStepSize. number	≥0	SHALL be used. The value of Element "number" SHALL be greater than or equal to zero.
7: O	valueStepSize. scale	≥0	MAY be used. If absent, a default value of "0" applies. If present, the value of Element "scale" SHALL be greater than or equal to zero.
9: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
9: M	measurementId	<m11#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
9: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
9: M	valueRangeMin. number		SHALL be used.
9: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
9: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
9: M	valueRangeMax. number		SHALL be used.
9: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
9: R	valueStepSize.		SHOULD be used. [Scaled number rules]
9: M	valueStepSize. number		SHALL be used.
9: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.

Table 44: Content of Function "measurementConstraintsListData" at Actor Battery

3.2.2.2.6.3 Function "measurementListData"

Scenario [{...}]: M/R/O [\W][\C]	**Element**	**Value**	**[High-Level Mapping] Element and value rules**
2: M	Measurement. measurementListData. measurementData.		
2: M	measurementId	<m1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#(1..1)->m1#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
2: M	value.		[MOB-002/2], [MOB-021] [Measurement value rules] [Scaled number rules]
2: M	value. number		SHALL be used.
2: O	value. scale		MAY be used. If absent, a default value of "0" applies.
2: M	valueSource	"calculatedValue"	
2: R	valueState		[MOB-005] [Value state rules]
2: M	Measurement. measurementListData. measurementData.		
2: M	measurementId	<m2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#(1..1)->m2#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
2: M	value.		[MOB-002/2], [MOB-022] [Measurement value rules] [Scaled number rules]
2: M	value. number		SHALL be used.
2: O	value. scale		MAY be used. If absent, a default value of "0" applies.
2: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
2: R	valueState		[MOB-005] [Value state rules]
2: M	Measurement. measurementListData. measurementData.		
2: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#(1..1)->m3#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.

2: M	value.		[MOB-002/3], [MOB-023] [Measurement value rules] [Scaled number rules]
2: M	value. number		SHALL be used.
2: O	value. scale		MAY be used. If absent, a default value of "0" applies.
2: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
2: R	valueState		[MOB-005] [Value state rules]
2: M	Measurement. measurementListData. measurementData.		
2: M	measurementId	<m4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#(1..1)->m4#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
2: M	value.		[MOB-002/3], [MOB-024] [Measurement value rules] [Scaled number rules]
2: M	value. number		SHALL be used.
2: O	value. scale		MAY be used. If absent, a default value of "0" applies.
2: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
2: R	valueState		[MOB-005] [Value state rules]
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m5#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#(1..1)->m5#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MOB-001], [MOB-031] [Measurement value rules] [Scaled number rules]
3: M	value. number		SHALL be used.
3: O	value. scale		MAY be used. If absent, a default value of "0" applies.
3: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
3: R	valueState		[MOB-005] [Value state rules]
4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m6#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.

4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#(1..1)->m6#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
4: M	value.		[MOB-001], [MOB-041] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: O	value. scale		MAY be used. If absent, a default value of "0" applies.
4: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: R	valueState		[MOB-005] [Value state rules]
5: M	Measurement. measurementListData. measurementData.		
5: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
5: O	timestamp	<t#(1..1)->m7#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
5: M	value.		[MOB-002/1], [MOB-051] [Measurement value rules] [Scaled number rules]
5: M	value. number		SHALL be used.
5: O	value. scale		MAY be used. If absent, a default value of "0" applies.
5: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
5: R	valueState		[MOB-005] [Value state rules]
6: M	Measurement. measurementListData. measurementData.		
6: M	measurementId	<m8#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
6: O	timestamp	<t#(1..1)->m8#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
6: M	value.		[MOB-001], [MOB-061] [Measurement value rules] [Scaled number rules]
6: M	value. number		SHALL be used.
6: O	value. scale		MAY be used. If absent, a default value of "0" applies.
6: O	evaluationPeriod.		[Evaluation period rules]
6: M	evaluationPeriod. startTime		
6: M	evaluationPeriod. endTime		

6: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
6: R	valueState		[MOB-005] [Value state rules]
6: M	Measurement. measurementListData. measurementData.		
6: M	measurementId	<m9#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
6: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
6: O	timestamp	<t#(1..1)->m9#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
6: M	value.		[MOB-001], [MOB-062] [Measurement value rules] [Scaled number rules]
6: M	value. number		SHALL be used.
6: O	value. scale		MAY be used. If absent, a default value of "0" applies.
6: O	evaluationPeriod.		[Evaluation period rules]
6: M	evaluationPeriod. startTime		
6: M	evaluationPeriod. endTime		
6: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
6: R	valueState		[MOB-005] [Value state rules]
7: M	Measurement. measurementListData. measurementData.		
7: M	measurementId	<m10#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
7: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
7: O	timestamp	<t#(1..1)->m10#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
7: M	value.		[MOB-071] [Measurement value rules] [Scaled number rules]
7: M	value. number	≥0	SHALL be used. The value of Element "number" SHALL be greater than or equal to zero.
7: O	value. scale	≥0	MAY be used. If absent, a default value of "0" applies. If present, the value of Element "scale" SHALL be greater than or equal to zero.
7: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
7: R	valueState		[MOB-005] [Value state rules]
9: M	Measurement. measurementListData. measurementData.		

9: M	measurementId	<m11#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
9: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
9: O	timestamp	<t#(1..1)->m11#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
9: M	value.		[MOB-091] [Measurement value rules] [Scaled number rules]
9: M	value. number		SHALL be used.
9: O	value. scale		MAY be used. If absent, a default value of "0" applies.
9: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
9: R	valueState		[MOB-005] [Value state rules]

Table 45: Content of Function "measurementListData" at Actor Battery

3.2.2.3 Client data - Specializations

As this Actor has only server functionality, no Specializations are described within this section.

3.3 Pre-Scenario communication

3.3.1 General information

The Pre-Scenario communication is needed if a client does not know the corresponding addresses on the server or if the required subscriptions or bindings are not active. In this case certain general communication mechanisms SHALL be used within SPINE:

a) Detailed discovery: allows to discover resource addresses.
b) Binding: allows to bind to resource address, which is frequently necessary to obtain write privileges.
c) Subscription: allows to subscribe to resource addresses, which is necessary to receive unsolicited notifications if a resource changes during runtime.

It is possible to combine those steps for multiple Scenarios or also multiple Use Cases:

- E.g. if multiple Scenarios in multiple Use Cases use the same Feature, only one subscription needs to occur.
- E.g. a complete detailed discovery or a subscription to the NodeManagement Feature needs to occur only once for all Use Cases.

Depending on which Entity, Feature and Functions are used within a Scenario the payload of the corresponding messages may slightly differ, but the basic principles and messages used stay the same.

The subsequent messages SHALL be exchanged for those parts that have not already been performed since the current connection is established or if those parts are outdated for another reason (e.g. if the detailed discovery is needed, but the bindings and subscriptions are still active from a previous connection only the detailed discovery messages need to be exchanged). If all Pre-Scenario communication parts are up-to-date, this section MAY be skipped, and the implementation can proceed as described in the corresponding "Scenario communication" sections.

After the connection is re-established (e.g. due to a power loss or a firmware update) the Pre-Scenario communication SHALL be performed as well. There may be circumstances where messages from the Pre-Scenario communication may be exchanged again.

Often the necessary messages of different Scenarios can be combined, so that only one single message is needed instead of multiple messages for the different Scenarios. This also is the case for the Pre-Scenario communication. In most cases only one "read" operation on the detailed discovery is necessary, as well as only one subscription request or binding request is needed for each Feature. Often multiple Scenarios within a Use Case access the same Feature, so only one subscription or binding is necessary.

3.3.2 Detailed discovery

For the functionality where a client already has current detailed discovery information (i.e. independent of this Use Case or any Scenario of it) the remainder of this section SHOULD be skipped.

Otherwise, the following procedure SHALL be performed in the given order:

1. If a client is not subscribed to the primary NodeManagement instance, the client SHALL acquire a subscription according to the figure provided within this sub-section.
2. A client SHALL read the detailed discovery information according to the figure provided within this sub-section. It SHALL keep the received information as far as it concerns mandatory and supported optional Entity Types, Feature Types and Functions of this Use Case that are needed by the client. This means that a client may choose how to store the necessary information. E.g. a client Actor can store the information how to address the necessary Features of the implemented Scenarios but may discard the Entity information.
3. If and as long as a client has a subscription to the detailed discovery information of an Actor and receives proper notifications, it SHALL consider (i.e. integrate into the kept detailed discovery information) the received information as far as it concerns mandatory and supported optional Entity Types, Feature Types and Functions of this Use Case.

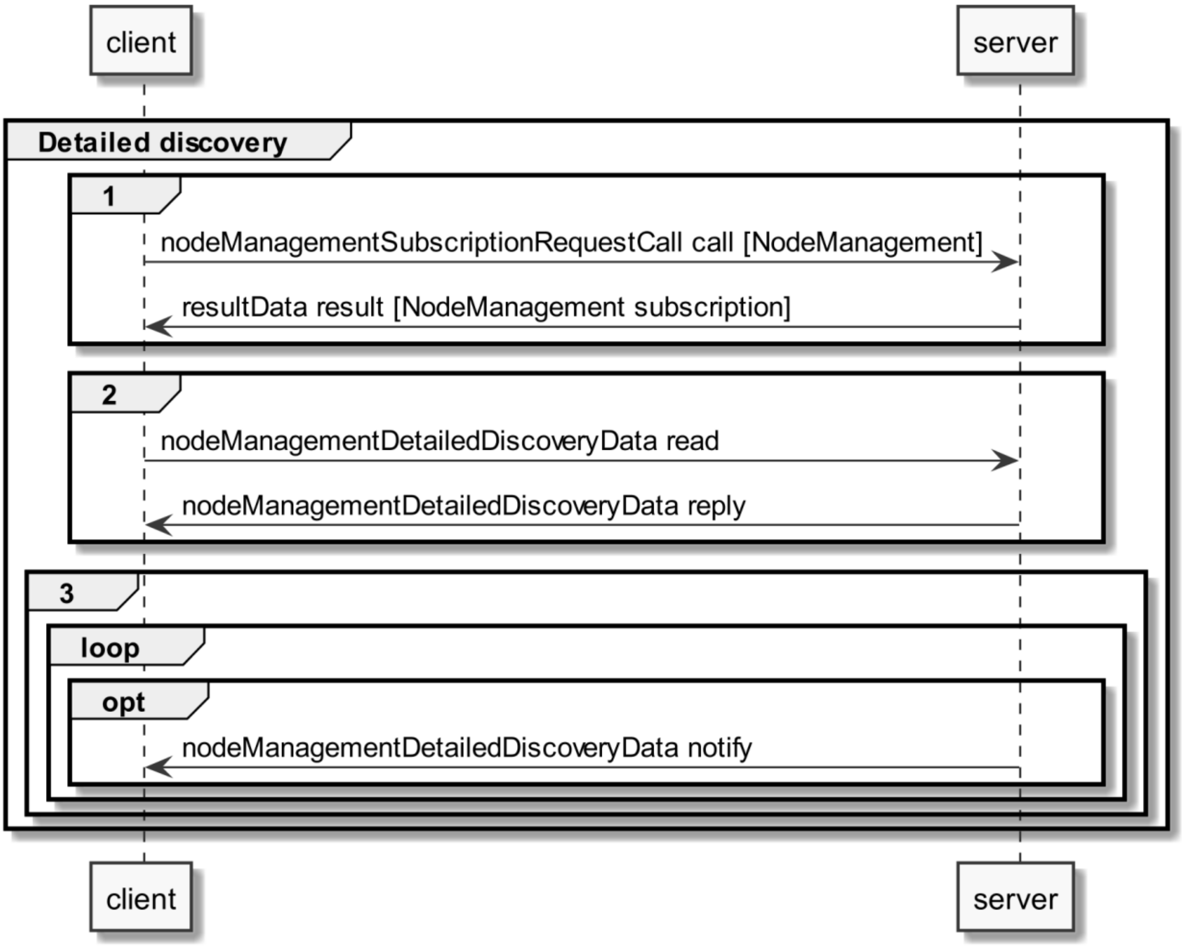


Figure 7: Pre-Scenario communication - Detailed discovery sequence diagram

If the "nodeManagementDetailedDiscoveryData read" fails, the client SHOULD retry to read the detailed discovery information until the "nodeManagementDetailedDiscoveryData reply" message was received successfully.

If all functionality is present at all times: The "nodeManagementDetailedDiscoveryData reply" message contains at least the mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as the used Functions and their "possible operations" described in section 3.2 and its sub-sections.

If functionality is added or removed dynamically: The "nodeManagementDetailedDiscoveryData reply" message does not need to contain all mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as all needed Functions and their "possible operations" described in section 3.2 and its sub-sections. However, as soon as the functionality is available it will be announced via a "nodeManagementDetailedDiscoveryData notify" message.

For the nodeManagementDetailedDiscoveryData read Function it is recommended to use a partial read with separated Selectors that may use one of the following Elements:

- entityType
- featureType

Note: Even with the usage of Selectors Features and Entities that are not relevant for this Use Case may be discovered. However, only Features and Entities that fulfil the hierarchical order as described within the Actors' sections shall be considered for this Use Case.

A "partial" notify SHALL be supported without using Selectors and Elements. Partial "delete" notify SHOULD also be supported with separated Selectors that may use one of the following Elements:

- entityAddress
- featureAddress

3.3.3 Binding

If binding is required by a Scenario that uses Features with writeable or changeable data, the server SHALL support binding for the respective Features. Before a write on a Function of a Feature occurs, the client SHALL create a binding to the corresponding Feature. For this the nodeManagementBindingRequestCall Function is used as shown in the following sequence diagram:

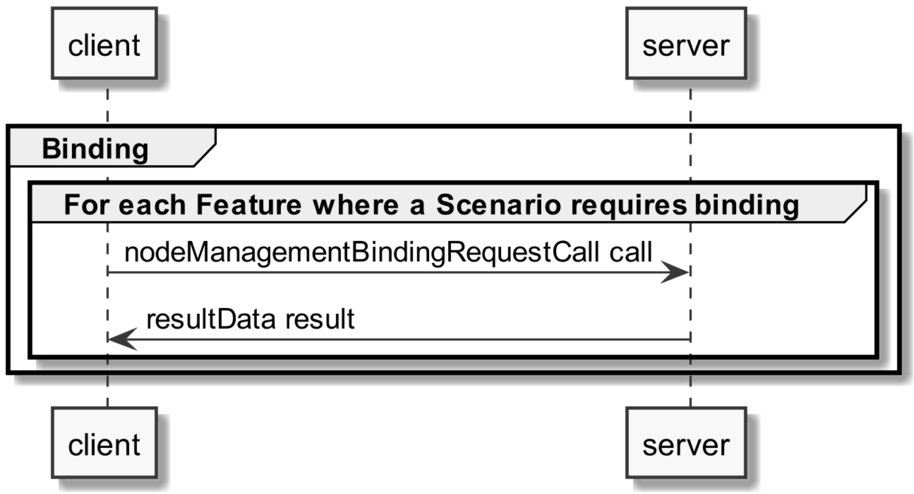


Figure 8: Pre-Scenario communication - Binding sequence diagram

If functionality is added or removed dynamically, binding may not be possible at all times on the required Functions. A client SHALL retry to create a binding again when receiving according updated detailed discovery information.

3.3.4 Subscription

A server SHALL support subscription for all Features that contain readable data that may change during runtime. The client SHALL create a subscription for all Features that the client wants to read. For this the nodeManagementSubscriptionRequestCall Function is used as shown in the following sequence diagram:

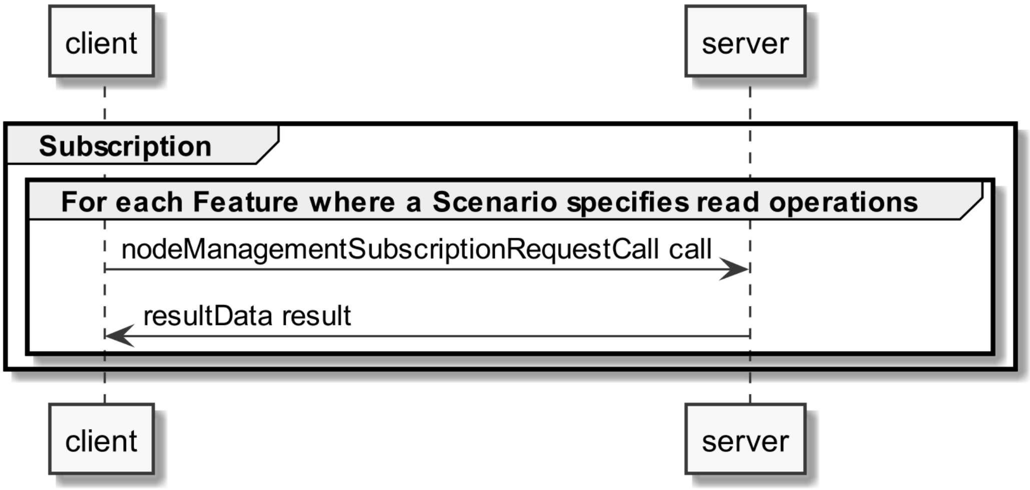


Figure 9: Pre-Scenario communication - Subscription sequence diagram

If the subscription request fails (e.g. because it is not supported by the server or the maximum number of possible subscriptions is reached), the client SHOULD read the data periodically (so-called "polling").

If functionality is added or removed dynamically, subscription may not be possible at all times on the required Functions. A client SHALL retry its subscription procedure again when receiving according updated detailed discovery information.

3.3.5 Dynamic behaviour

In case Entities or Features are removed, a nodeManagementDetailedDiscoveryData "notify" is transmitted that informs about the deleted Entities and Features. All existing binding or subscription entries on the deleted Features SHALL be deleted by each device.

In case Entities or Features are added the Pre-Scenario communication starts with transmitting a nodeManagementDetailedDiscoveryData "notify" that contains the added Entities and Features.

3.4 Scenarios

3.4.1 Scenario 1-9 - Overall Scenario information

3.4.1.1 Pre-Scenario communication

1. **Detailed Discovery**: Actors that act as client within the Scenarios of this Use Case, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding**: Binding SHOULD NOT be used for the Scenarios of this Use Case.
3. **Subscription**: Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within the Scenarios of this Use Case, as described in section 3.3.4.

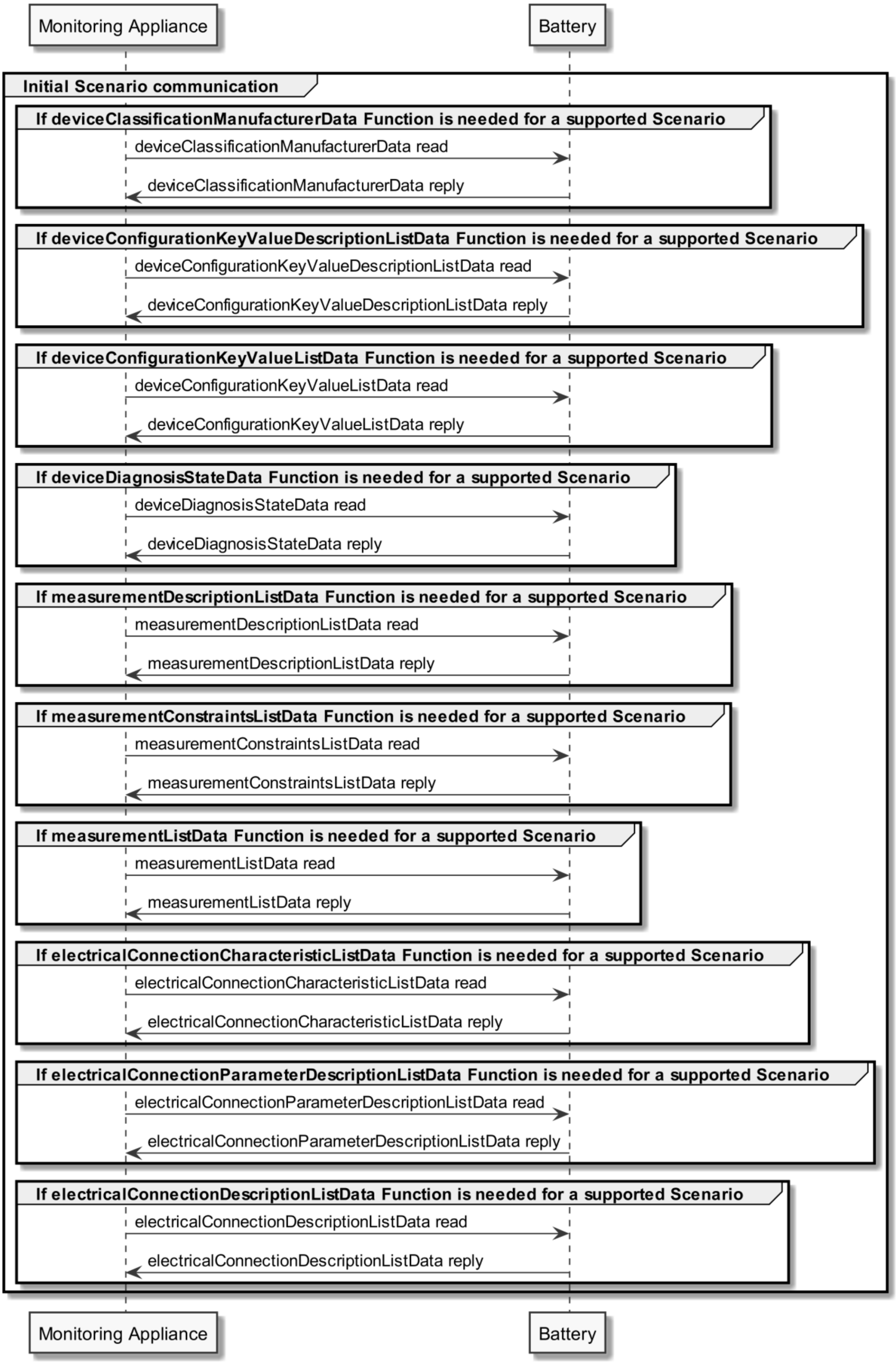
The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

Note: This section holds for all Scenarios of this Use Case.

3.4.1.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the following messages as shown in the sequence diagram SHALL be exchanged (if a supported Scenarios requires the respective messsage), as the corresponding resources may have changed in the meantime:



1250

1251 *Figure 10: Scenario 1-9 - Initial Scenario communication sequence diagram*

The following table shows where the required content of the messages from the sequence diagram is described:

Message name from sequence diagram	**Content description in table**	**Relevant for Scenario**
deviceClassificationManufacturerData reply	Table 36	1
deviceConfigurationKeyValueDescriptionListData reply	Table 37	7
		8
deviceConfigurationKeyValueListData reply	Table 38	7
		8
deviceDiagnosisStateData reply	Table 39	2
electricalConnectionDescriptionListData reply	Table 40	2
		3
		4
		5
		6
electricalConnectionParameterDescriptionListData reply	Table 41	2
		3
		4
		5
		6
electricalConnectionCharacteristicListData reply	Table 42	8
measurementDescriptionListData reply	Table 43	2
		3
		4
		5
		6
		7
		9
measurementConstraintsListData reply	Table 44	2
		3
		4
		5
		6
		7
		9
measurementListData reply	Table 45	2
		3
		4
		5
		6
		7
		9

Table 46: Initial Scenario communication content references for Scenario 1-9

Note: Within the Initial Scenario communication, the content required by the Scenarios of this Use Case MAY not be provided completely, but later during Runtime Scenario communication.

Note: This section holds for all Scenarios of this Use Case.

3.4.1.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

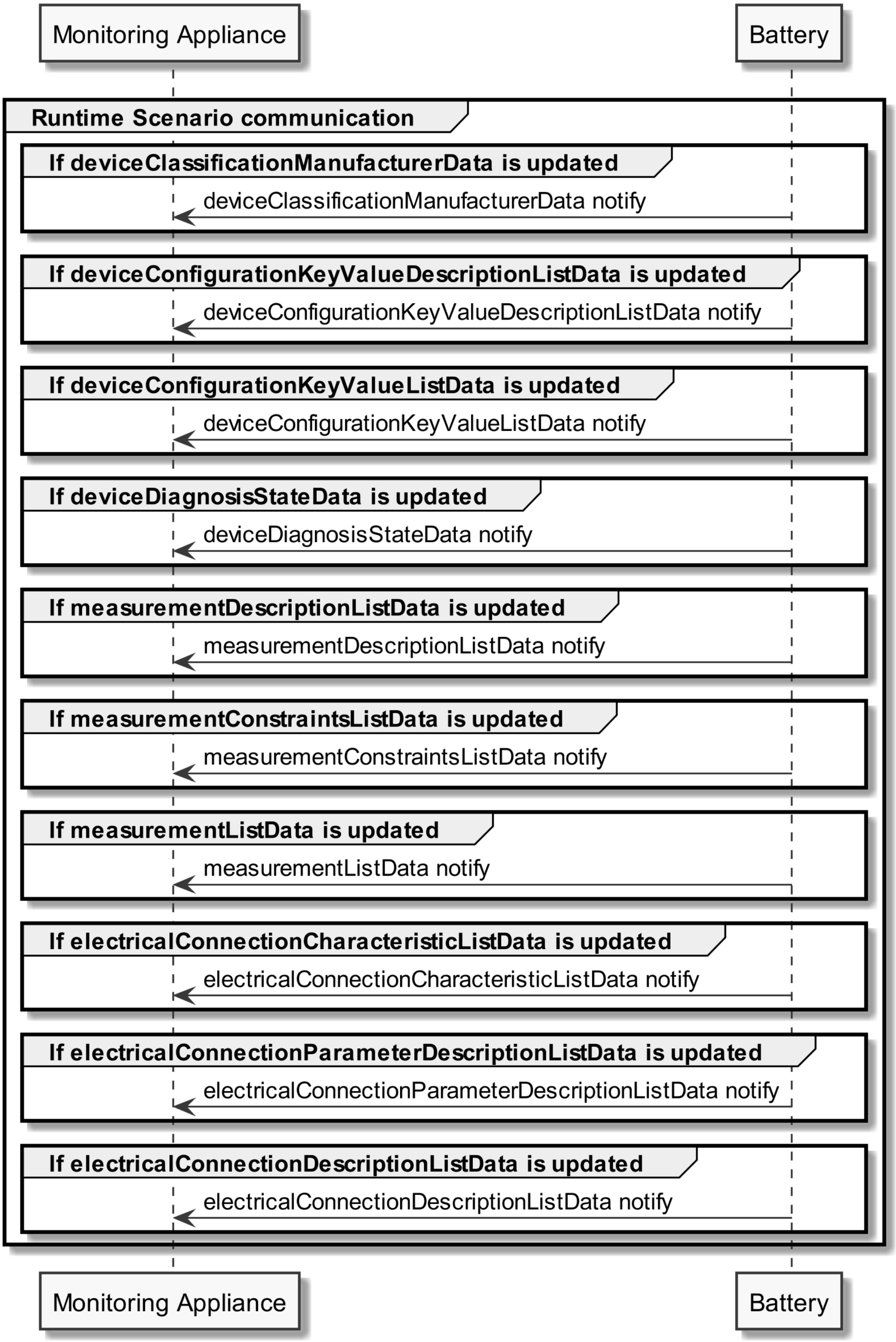


Figure 11: Scenario 1-9 - Runtime Scenario communication sequence diagram

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	**Content description in table**	**Relevant for Scenario**
deviceClassificationManufacturerData notify	Table 36	1
deviceConfigurationKeyValueDescriptionListData notify	Table 37	7
		8
deviceConfigurationKeyValueListData notify	Table 38	7
		8
deviceDiagnosisStateData notify	Table 39	2
electricalConnectionDescriptionListData notify	Table 40	2
		3
		4
		5
		6
electricalConnectionParameterDescriptionListData notify	Table 41	2
		3
		4
		5
		6
electricalConnectionCharacteristicListData notify	Table 42	8
measurementDescriptionListData notify	Table 43	2
		3
		4
		5
		6
		7
		9
measurementConstraintsListData notify	Table 44	2
		3
		4
		5
		6
		7
		9
measurementListData notify	Table 45	2
		3
		4
		5
		6
		7
		9

Table 47: Runtime Scenario communication content references for Scenario 1-9

Note: This section holds for all Scenarios of this Use Case.

3.4.2 Scenario 1 - Monitor Battery identification

3.4.2.1 Additional information

None.

3.4.3 Scenario 2 - Monitor Battery state

3.4.3.1 Additional information

None.

3.4.3.2 Information access

Partial data information regarding [MOB-021]:

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "stateOfCharge"

The measurementConstraintsListData read and measurementListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [MOB-022]:

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "stateOfHealth"

The measurementConstraintsListData read and measurementListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [MOB-023]:

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "stateOfEnergy"

The measurementConstraintsListData read and measurementListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [MOB-024]:

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "useableCapacity"

The measurementConstraintsListData read, measurementListData and electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

3.4.4 Scenario 3 - Monitor Battery DC power

3.4.4.1 Additional information

None.

3.4.4.2 Information access

Partial data information regarding [MOB-031]:

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "dcPower"

The measurementConstraintsListData read, measurementListData read and electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

3.4.5 Scenario 4 - Monitor Battery DC current

3.4.5.1 Additional information

None.

3.4.5.2 Information access

Partial data information regarding [MOB-041]:

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "dcCurrent"

The measurementConstraintsListData read, measurementListData read and electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

3.4.6 Scenario 5 - Monitor Battery DC voltage

3.4.6.1 Additional information

None.

3.4.6.2 Information access

Partial data information regarding [MOB-051]:

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "dcVoltage"

The measurementConstraintsListData read, measurementListData read and electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

3.4.7 Scenario 6 - Monitor Battery DC energy

3.4.7.1 Additional information

None.

3.4.7.2 Information access

Partial data information regarding [MOB-061]:

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "dcChargeEnergy"

The measurementConstraintsListData read, measurementListData read and electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [MOB-062]:

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "dcDischargeEnergy"

The measurementConstraintsListData read, measurementListData read and electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

3.4.8 Scenario 7 - Monitor Battery additional details

3.4.8.1 Additional information

Note on [MOB-071]: As the Cumulated Load Cycle Count ([MOB-071]) is an integer value, its numerical representation via the Elements "number" and "scale" is restricted: Both Elements SHALL be greater than or equal to zero (see Table 45 and also Table 44). This also means that representations like { number=200, scale=-2 } are forbidden, even though the calculation of

$$\text{number} * 10^{\text{scale}}$$

(see [Scaled number rules]) would yield an integer value of "2" in this example.

3.4.8.2 Information access

Partial data information regarding [MOB-071]:

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "loadCycleCount"

The measurementConstraintsListData read and measurementListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [MOB-072]:

The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- keyName = "batteryType"

The deviceConfigurationKeyValueListData read SHOULD be a "partial" read operation with the following Selector:

- keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

3.4.9 Scenario 8 - Monitor Battery capabilities

3.4.9.1 Additional information

Note on [MOB-084]: As the Battery Maximum Charge/Discharge Cycle Count Per Day ([MOB-084]) is an integer value, its numerical representation via the Elements "number" and "scale" is restricted: Both Elements SHALL be greater than or equal to zero (see Table 38). This also means that representations like { number=200, scale=-2 } are forbidden, even though the calculation of

number * 10^{scale}

(see [Scaled number rules]) would yield an integer value of "2" in this example.

3.4.9.2 Information access

Partial data information regarding [MOB-081]:

The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the following Selector:

- electricalConnectionId = <if [MOB-024] is supported by Actor Monitoring Appliance AND available on Actor Battery, the same value as used within [MOB-024] SHALL be used here; otherwise, the electricalConnectionId is not needed within the "partial" read operation>
- parameterId = <if [MOB-024] is supported by Actor Monitoring Appliance AND available on Actor Battery, the same value as used within [MOB-024] SHALL be used here; otherwise, the parameterId is not needed within the "partial" read operation>
- characteristicContext = "entity"
- characteristicType = "energyCapacityNominalMax"

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [MOB-082]:

The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the following Selector:

- electricalConnectionId = <same value as used within [MOB-031]>
- parameterId = <same value as used within [MOB-031]>
- characteristicContext = "entity"
- characteristicType = "powerConsumptionNominalMax"

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [MOB-083]:

The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the following Selector:

- electricalConnectionId = <same value as used within [MOB-031]>
- parameterId = <same value as used within [MOB-031]>
- characteristicContext = "entity"
- characteristicType = "powerProductionNominalMax"

Note: If partial read is not supported, a full read SHALL be performed.

Partial data information regarding [MOB-084]:

The deviceConfigurationKeyValueDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- keyName = "maxCycleCountPerDay"

The deviceConfigurationKeyValueListData read SHOULD be a "partial" read operation with the following Selector:

- keyId (derived from the deviceConfigurationKeyValueDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

3.4.10 Scenario 9 - Monitor Internal data

3.4.10.1 Additional information

None.

3.4.10.2 Information access

Partial data information regarding [MOB-091]:

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "componentTemperature"

The measurementConstraintsListData read and measurementListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

